Causal Inference in Observational Studies

Contents

Disclaimers: (i) Sections and lines in brown are under construction. (ii) For all mathematical simplifications featuring "= ... =", the detailed steps are provided in the Appendix.

1 Definitions

A research design is the overall plan, given a research question, for collecting and analyzing data in order to draw credible conclusions.

Let us set the scene for the present document. We are concerned with the branch of econometrics that seeks to identify *causal* effects, particularly from observational data. This is a real challenge: relationships between observed variables are easy to estimate, but how do we know whether they are causal or spurious?

Precisely, we want to estimate the causal effect of an intervention or *treatment* for some population. We need a research design that is able to credibly *identify* the effect. A central component of the research design of causal inference studies is hence the *identification strategy.*[1]

An identification strategy is the argument of how observational data are used to recover the causal effect of interest. It specifies notably:

- a source of identifying variation in the treatment variable,
- the assumptions under which this variation identifies the causal effect.

An estimation strategy then specifies the econometric methods used to estimate the identified effect.

Different experiments will provide different types of variation in the treatment. For causal inference, when internal validity is the priority,[2] a research design based on a randomized trial is commonly regarded as the gold standard. By randomly assigning treatment across units, a randomized trial prevents systematic selection into treatment and makes the treated and the untreated groups comparable in expectation, thereby allowing treatment effects to be identified under the relevant assumptions—this will be detailed in later sections. As observational studies strive to achieve this strength of evidence, their identification strategies generally seek to approximate such an experiment.

Experiments

A true experiment is a study in which the researcher actively assigns or manipulates the treatment and observes its effect on an outcome. In a **randomized experiment**, treatment is randomly assigned, making treatment status independent of potential outcomes in expectation.

A natural experiment is an observational study in which plausibly exogenous[a]—often *as-if random*—variation in a treatment D or an instrument Z has occurred—mimicking the exogeneity of a randomized experiment. Researchers don't create natural experiments, they find them.

Ex: Certain weather events, natural disasters, lotteries, or arbitrary assignment rules.

A quasi-experiment[b] is a study of intentional treatment that resembles a randomized experiment but lacks full researcher-controlled random assignment. Because the treated group and the untreated group (often called *comparison* group rather than *control*) may differ systematically, causal identification relies on the study's comparison structure, design features, and specific identifying assumptions.

Ex: In the 1990s, the U.S. Department of Housing and Urban Development (HUD) implemented a grant program to encourage resident management of low-income public housing projects. Housing projects were *selected* in 11 cities nationwide, so the treatment (the award of HUD funding) was not randomly assigned. But similar housing projects in the same cities provided a reasonably valid comparison, so the HUD was able to evaluate the program.

[a] Exogeneity is w.r.t. potential outcomes, i.e., the variation is unrelated to the factors that would otherwise determine the outcome.

[b] ⚠ Terminology varies across disciplines; econometricians often use the term 'quasi-experimental' to refer to any credible nonrandomized design—including natural experiments.

[1] Angrist and Pischke (2008) uses the notions of research design and identification strategy interchangeably.

[2] The notions of internal vs. external validity are defined in Section ??. Let us note for the time being that the methods of choice for internal validity may atually limit external validity. For example, a zoo is a controllable setting amenable to drawing causal inferences about animal behavior, but those inferences may not generalize to animals in the wild.

2 Framework: a counterfactual approach to causality

2.1 The potential outcomes framework

2.1.1 The original selection bias problem and independence assumptions

Consider a population, of which we observe a sample of units $i = 1, \ldots, n$, and a binary treatment $D_i \in \{0, 1\}$ whose causal effect on an outcome Y we want to estimate. Let Y_i^1 denote unit i's potential outcome[3] if they were to receive the treatment, Y_i^0 otherwise, and Y_i their actual realized outcome. We assume that there is no interference between units.[4] The potential outcomes framework allows for defining estimands:

Individual treatment effects (TEs)	$Y_i^1 - Y_i^0,\ \forall i$	*what we would ideally estimate*
Average treatment effect (ATE)	$\mathbb{E}\left[Y_i^1 - Y_i^0\right]$	*what we reasonably want to estimate*
Average treatment effect on the treated (ATET)	$\mathbb{E}\left[Y_i^1 - Y_i^0 \mid D_i{=}1\right]$	"
Conditional average treatment effect (CATE)	$\mathbb{E}\left[Y_i^1 - Y_i^0 \mid \mathrm{X}_i\right]$	"
Difference in mean observed outcomes	$\mathbb{E}[Y_i \mid D_i{=}1] - \mathbb{E}[Y_i \mid D_i{=}0]$	*what we can estimate*
All expectations can be defined conditional on covariates X*, yielding then conditional means within strata.*		

The original selection bias problem To measure $\mathrm{TE}_i = Y_i^1 - Y_i^0$, we need to observe the same individual with and without treatment. This is impossible, we can never observe the counterfactual.[5] What we can only estimate is the difference in means of observed outcomes, but this quantity contains a selection bias, stemming from the difference in mean Y_i^0 between the treated and the untreated:[6]

$$\mathbb{E}[Y_i \mid D_i{=}1] - \mathbb{E}[Y_i \mid D_i{=}0] \;=\; \ldots \;=\; \underbrace{\mathbb{E}\left[Y_i^1 - Y_i^0 \mid D_i{=}1\right]}_{\text{ATET}} + \underbrace{\mathbb{E}\left[Y_i^0 \mid D_i{=}1\right] - \mathbb{E}\left[Y_i^0 \mid D_i{=}0\right]}_{\text{selection bias}}$$

Identification, i.e., elimination of the selection bias, can be achieved through analysis, depending on the assumed *treatment assignment* mechanism.[7] We distinguish three classes of assignment mechanisms:

- *Random assignment*

  If treatment is randomly assigned, it is independent of potential outcomes: $D_i \perp\!\!\!\perp (Y_i^0, Y_i^1)$, so there is no selection bias *in expectation*.[8] Since independence also implies ATET=ATE, this independence assumption (IA) identifies the ATE.

- *Selection on observables*

  Generally, in observational studies, $D_i \not\!\perp\!\!\!\perp (Y_i^0, Y_i^1)$. However, if we can *match* treated and untreated units to be proper counterfactuals, i.e., if conditional on some pre-treatment characteristics X_i, treatment assignment is independent of potential outcomes: $D_i \perp\!\!\!\perp (Y_i^0, Y_i^1) \mid \mathrm{X}_i$, and if there are both treated and control units across the support of X_i (i.e., overlap), then we can again eliminate selection bias in expectation. We compare outcomes *for given* X_i, i.e., within stratum:

$$\underbrace{\mathbb{E}[Y_i \mid D_i{=}1, \mathrm{X}_i] - \mathbb{E}[Y_i \mid D_i{=}0, \mathrm{X}_i]}_{\text{difference in mean outcomes given } \mathrm{X}_i} \;=\; \ldots \;=\; \underbrace{\mathbb{E}\left[Y_i^1 - Y_i^0 \mid D_i{=}1, \mathrm{X}_i\right]}_{\text{ATET given } \mathrm{X}_i\text{: CATET}} + \underbrace{\mathbb{E}\left[Y_i^0 \mid D_i{=}1, \mathrm{X}_i\right] - \mathbb{E}\left[Y_i^0 \mid D_i{=}0, \mathrm{X}_i\right]}_{\text{selection bias given } \mathrm{X}_i}$$

[3] The potential outcomes framework for causal inference builds on Neyman (1923), was extended to observational studies by Rubin (1974), and became popular in econometrics around 1990.

[4] The Stable Unit Treatment Value Assumption (SUTVA) states that the treatment received by one unit does not affect the potential outcomes of other units. This notably excludes spillovers and strategic interactions. When this is unrealistic, alternative approaches include redefining the unit of interest (aggregating), or directly modeling the interactions.

[5] This is the *fundamental problem of causal inference*. Its implication: we never *observe* causal effects.

[6] The selection bias is the sum of the net average balance (i.e., the net difference of means) of other causes across the two groups. For example, if individuals with low Y_i^0 choose treatment more frequently, then $\mathbb{E}\left[Y_i^0 \mid D_i{=}1\right] < \mathbb{E}\left[Y_i^0 \mid D_i{=}0\right]$; comparing Y between treated and untreated underestimates the TE. Say we look at the effect of hospitalization; sick individuals go to the hospital (get treated) more often than healthy individuals. But they would also have been less healthy had they stayed at home.

[7] Imbens and Wooldridge (2009) formally defines the assignment mechanism as $\Pr(D_i{=}1 \mid Y_i^0, Y_i^1, \mathrm{X}_i)$.

[8] Independence removes selection bias *in this expectation form*, where the expectation is taken over repeated randomizations on the trial sample, each with its own allocation of treatments and controls (Deaton and Cartwright, 2018). Independence does not imply actual balance in any single trial. Therefore in any RCT, the sample analog of the last term may not actually be 0.

The conditional independence assumption (CIA) eliminates selection bias, and also implies CATET = CATE, thereby identifying the CATEs. We may then combine these CATEs by weighting them in some way to recover a single weighted average.[9]

- *Selection on unobservables*

  We can't assume that we observe all factors that are associated with both the assignment and the potential outcomes, i.e., the CIA fails. To identify some treatment effect(s), we need other identification strategies whose design-specific assumptions will identify design-specific estimands, often more local than the ATE. The most common methods are described in Section 3.

> The credible identification of treatment effects requires **identifying assumptions** that make a counterfactual comparison valid.
>
> - Independence assumptions[10] are the strongest:
>   - The IA identifies the ATE.
>   - The {CIA + overlap} identify CATEs—and from there, chosen aggregates.
> - Alternative assumptions may also yield identification:
>   - *Instrumental variables* – {An exogenous relevant instrument Z for D + exclusion and monotonicity} identify the LATE for compliers.
>   - *Regression discontinuity* – Continuity of the potential-outcome conditional means at a treatment cutoff identifies a LATE at the cutoff.
>   - *Difference-in-differences* – {Parallel counterfactual trends + no anticipation} identify an ATET for a specified group and period.

Beyond average effects: distributional effects The identification achieved through independence assumptions extends beyond average treatment effects. Independence implies that the full marginal distribution of each potential outcome is identified. We can hence identify quantile treatment effects (QTEs), which are the effects of the treatment at corresponding quantiles of the two marginal potential-outcome distributions: $\tau_p := \mathbb{Q}_{Y^1}(p) - \mathbb{Q}_{Y^0}(p),\ \forall p \in [0,1]$.[11] QTEs can reveal distributional treatment effects obscured by the ATE, e.g., effects concentrated in the tails of the outcome distribution.

Note that under the CIA, one identifies *conditional* quantile treatment effects, $\tau_p(\mathrm{x}) := \mathbb{Q}_{Y^1|\mathrm{X}=\mathrm{x}}(p) - \mathbb{Q}_{Y^0|\mathrm{X}=\mathrm{x}}(p)$. However, in general, the unconditional QTE $\tau_p \neq \mathbb{E}_{\mathrm{X}}[\tau_p(\mathrm{X})]$. Consequently, τ_p cannot generally be obtained by averaging conditional QTEs. One must instead average the conditional distribution functions over X and then invert the resulting marginal distribution functions (Imbens and Wooldridge, 2009).

2.1.2 Average treatment effects as regression slopes and endogeneity

Let $\beta_i := Y_i^1 - Y_i^0$ denote the individual treatment effect for unit i. We distinguish a causal regression-form decomposition from the population OLS projection:

- *Causal regression-form decomposition:* The relation between observed and potential outcomes can be

[9] For instance, the matching estimand will weight them by the distribution of X among the treated, which recovers the ATET, whereas the linear regression estimand will weight them by the conditional variance of D, emphasizing strata with substantial treatment-control overlap—see Section 2.1.2.

[10] Independence assumptions are also called *unconfoundedness* or *ignorability* assumptions in statistics, meaning ignorability of the assignment mechanism. Indeed, with independence, we don't need to know and model the assignment process to estimate causal effects, we need only compare group means. Earlier, we described the assignment mechanisms of random assignment and selection on observables solely for they imply what we are actually interested in for identification: the IA and the CIA, respectively.

[11] ⚠ The p-th QTE $\mathbb{Q}_{Y^1}(p) - \mathbb{Q}_{Y^0}(p)$ is the effect of the treatment at quantile p of the two marginal potential outcome distributions. It is not the p–th quantile of the individual treatment effect $\mathbb{Q}_{Y^0 - Y^1}(p)$. In general, the latter quantile of a difference differs from the difference in quantiles. Independence identifies the marginal distributions of Y^0 and Y^1, but not their joint distribution and hence not the distribution of $Y^1 - Y^0$. There is no similar problem in estimating the ATE, as $E[Y^1 - Y^0] = \mathbb{E}[Y^1] - \mathbb{E}[Y^0]$: differences in means equal means of differences.

rewritten through the exact algebraic causal decomposition

$$
\begin{aligned}
Y_i &= Y_i^0 + \beta_i D_i \\
&= Y_i^0 + \beta_i D_i + \mathbb{E}[Y_i^0 \,|\, D_i{=}0] - \mathbb{E}[Y_i^0 \,|\, D_i{=}0] + \mathbb{E}[\beta_i \,|\, D_i{=}1] D_i - \mathbb{E}[\beta_i \,|\, D_i{=}1] D_i \\
&= \underbrace{\mathbb{E}[Y_i^0 \,|\, D_i{=}0]}_{:=\alpha^c} \;+\; \underbrace{\mathbb{E}[\beta_i \,|\, D_i{=}1]}_{:=\beta^c=\text{ATET}} D_i \;+\; \underbrace{Y_i^0 - \mathbb{E}[Y_i^0 \,|\, D_i{=}0] + (\beta_i - \mathbb{E}[\beta_i \,|\, D_i{=}1])\, D_i}_{:=u_i^c}
\end{aligned}
$$

The slope is the ATET, and the causal or structural disturbance u_i^c has conditional mean zero iff there is no selection bias—i.e., no imbalance in potential outcomes across the treatment groups—as we find that $\mathbb{E}[u_i^c \,|\, D_i{=}1] - \mathbb{E}[u_i^c \,|\, D_i{=}0] = \ldots = \underbrace{\mathbb{E}[Y_i^0 \,|\, D_i{=}1] - \mathbb{E}[Y_i^0 \,|\, D_i{=}0]}_{\text{selection bias}}$.

Endogeneity In a regression, a regressor is called *exogenous* if it is uncorrelated with the disturbance and *endogenous* otherwise. Thus, here, selection bias appears as endogeneity of D_i: $\text{cov}[D_i, u_i^c] \neq 0$.

Sources of endogeneity include:

- *Reverse causality or simultaneity:* if Y also affects D, determinants of Y contained in u^c may be correlated with D;
- *Measurement error in D* that is correlated with Y;
- *Omitted variable bias (OVB):* all omitted variables[12] are captured by u^c. Therefore, if an omitted variable is correlated with D, u^c is correlated with D. Such a variable is called a *confounding* variable. This source of endogeneity is the most common.

- *Population OLS projection:*

$$
\begin{aligned}
Y_i &= \alpha_{\text{OLS}} + \beta_{\text{OLS}} D_i + e_i, \qquad \text{with } \mathbb{E}[e_i] = 0, \quad \text{cov}[D_i, e_i] = 0 \\
&= \mathbb{E}[Y_i \,|\, D_i{=}0] + \frac{\text{cov}[Y_i, D_i]}{\mathbb{V}[D_i]} D_i + e_i \\
&= \mathbb{E}[Y_i \,|\, D_i{=}0] + \ldots D_i + e_i \\
&= \mathbb{E}[Y_i \,|\, D_i{=}0] + (\mathbb{E}[Y_i \,|\, D_i{=}1] - \mathbb{E}[Y_i \,|\, D_i{=}0])\, D_i + e_i
\end{aligned}
$$

Its slope can be related to the causal quantities:

$$
\begin{aligned}
\beta_{\text{OLS}} &= \mathbb{E}[Y_i \mid D_i{=}1] - \mathbb{E}[Y_i \mid D_i{=}0] \\
&= \mathbb{E}[Y_i^1 \mid D_i{=}1] - \mathbb{E}[Y_i^0 \mid D_i{=}0] \\
&= \mathbb{E}[Y_i^1 \mid D_i{=}1] - \mathbb{E}[Y_i^0 \mid D_i{=}0] - \mathbb{E}[Y_i^0 \mid D_i{=}1] + \mathbb{E}[Y_i^0 \mid D_i{=}1] \\
&= \underbrace{\mathbb{E}[Y_i^1 - Y_i^0 \mid D_i{=}1]}_{\text{ATET}} + \underbrace{\mathbb{E}[Y_i^0 \mid D_i{=}1] - \mathbb{E}[Y_i^0 \mid D_i{=}0]}_{\text{selection bias}}.
\end{aligned}
$$

The two intercepts are always equal: $\alpha_{\text{OLS}} = \mathbb{E}[Y_i \,|\, D_i{=}0] = \mathbb{E}[Y_i^0 \,|\, D_i{=}0] = \alpha^c$, and the two slopes are equal iff there is no selection bias, or equivalently, iff $\mathbb{E}[u_i^c \,|\, D_i{=}1] = \mathbb{E}[u_i^c \,|\, D_i{=}0]$, i.e., the treatment is uncorrelated with the causal disturbance. Hence, under this condition, the causal regression-form decomposition coincides with the population OLS projection: OLS identifies the ATET![13]

> → In the simple setting of a binary D and no covariates,
>
> the population OLS slope identifies the ATET iff there is no selection bias
> iff D_i is exogenous w.r.t. u_i^c
>
> In observational studies, we can rarely be certain that conditioning on the observed covariates eliminates all confounding,[14] and so often risk selection bias... We hence turn to alternative identification strategies that rely on other identifying assumptions.

[12] An omitted variable is an explanatory variable not included in the regression but which is a determinant of Y.

[13] The absence of selection bias is sufficient for OLS to identify the ATET, but not necessarily the ATE. To additionally obtain $\beta_{\text{OLS}} = \text{ATE} := \mathbb{E}[Y_i^1 - Y_i^0]$, one must also have $\mathbb{E}[Y_i^1 - Y_i^0 \,|\, D_i{=}1] = \mathbb{E}[Y_i^1 - Y_i^0]$, so that the ATET equals the ATE. Full independence, $D_i \perp\!\!\!\perp (Y_i^0, Y_i^1)$, is a sufficient condition for both requirements and therefore implies $\beta_{\text{OLS}} = \text{ATET} = \text{ATE}$.

In more general settings, what treatment effect does the population OLS estimand recover?

We saw that when D is binary and there are no covariates X, (a) the population OLS slope equals the difference in observed conditional means: $\beta_{\text{OLS}} = \mathbb{E}[Y_i \,|\, D_i = 1] - \mathbb{E}[Y_i \,|\, D_i = 0]$, and (b) this in turn equals the ATET iff there is no selection bias. However, in more general settings, the population OLS estimand may diverge from the causal estimand of interest.

- **With covariates X**

  Treatment effects are simply differences in means. We could hence estimate them nonparametrically by comparing treated and untreated observations that share the same covariate values, as in exact matching. In practice, as the number of covariates increases, such nonparametric comparisons become increasingly difficult, so we use regression as a computational device. However, there are two main differences between the matching and the linear regression estimands:

  1. An unsaturated (in X) linear regression imposes functional-form assumptions on the conditional expectations of the potential outcomes given X. In particular, it generally relies on a global linear approximation to the regression functions. Even if linearity is locally reasonable, a poor global approximation may produce substantial specification bias (Imbens and Wooldridge, 2009). It may therefore be preferable to combine regression with matching or weighting, so that regression is used only to adjust for remaining differences between relatively comparable observations.

  2. Matching and OLS use different weights to aggregate the within-stratum differences in mean observed outcome $\delta_{\text{x}} := \mathbb{E}[Y_i \,|\, D_i{=}1, \text{X}_i{=}\text{x}] - \mathbb{E}[Y_i \,|\, D_i{=}0, \text{X}_i{=}\text{x}]$—which, under the CIA, identify the CATEs $\mathbb{E}[Y_i^1 - Y_i^0 \,|\, \text{X}_i{=}\text{x}]$. If δ_{x} varies with x, i.e., if the TE is heterogeneous w.r.t. X, then the different sets of weights will result in different weighted averages of treatment effects (Angrist and Pischke, 2008, 3.3.1). For simplicity, consider a univariate and discrete X_i.

     - In the exact-matching estimand, each stratum is weighted according to the conditional probability of treatment, which identifies the ATET.

       $$\beta_{\text{M}} = \ldots = \frac{\sum_x \delta_x w_{\text{M}}(x)}{\sum_x w_{\text{M}}(x)}, \qquad w_{\text{M}}(x) := \Pr(D_i{=}1 \,|\, X_i{=}x)\, \Pr(X_i{=}x)$$

     - In saturated OLS, each stratum is weighted according to the conditional variance of treatment, which is maximized when $\Pr(D_i{=}1|X_i) = .5$, i.e., in strata with as many treated as control observations: OLS gives more weight to more balanced strata.

       $$\beta_{\text{OLS}} = \ldots = \frac{\sum_x \delta_x w_{\text{R}}(x)}{\sum_x w_{\text{R}}(x)}, \quad w_{\text{R}}(x) := \Pr(D_i{=}1 \mid X_i{=}x)\, \big(1 - \Pr(D_i{=}1 \mid X_i{=}x)\big)\, \Pr(X_i{=}x)$$

  Hence, with heterogeneous treatment effects, the saturated OLS coefficient generally differs from the ATET. With an unsaturated linear specification for X, the OLS coefficient may additionally be affected by functional-form misspecification.

- **With a nonbinary *D***

  We consider a treatment with multiple levels of intensity. Let $D_i \in \{d^l, \ldots, d^u\}$ denote treatment intensity, and $Y_i(d)$ denote individual i's potential outcome under treatment level d—a dose-response function of d—and $Y_i := Y_i(D_i)$ their realized outcome. The generalized CIA and overlap conditions are $Y_i(d) \perp\!\!\!\perp D_i \mid \text{X}_i$ and $\Pr(D_i{=}d \mid \text{X}_i{=}\text{x}) > 0$ for all relevant values of (d, x). For example, the within-stratum CATET for a 1-unit increase in treatment is then $\delta_{\text{x}} := \mathbb{E}[Y_i(d) - Y_i(d{-}1) \mid D_i{=}d, \text{X}_i{=}\text{x}]$.

  - *Ideal case:* the dose-response function is linear in d and its slope does not vary with i: $\mathbb{E}[Y_i(d) \mid \text{X}_i] = \alpha + \beta d + \text{X}_i'\gamma$. Then a one-unit increase in treatment has the same causal effect β at every treatment level and for every x: $\mathbb{E}[Y_i(d) - Y_i(d-1) \mid \text{X}_i = \text{x}] = \beta$. Under the generalized CIA, the coefficient on D_i in the correctly specified regression $Y_i = \alpha + \beta D_i + \text{X}_i'\gamma + u_i$ identifies this common causal effect.

  - *General case:* the dose-response function is nonlinear in d or its slope varies across i or X. Then the above regression estimates a weighted average of differences in the average dose–response function $Y_i(d) - Y_i(d{-}1)$, but it does not generally equal a particular ATET. One may instead include indicators for the treatment levels or estimate the dose–response function flexibly.

- **With a limited Y**

  We consider an outcome that isn't continuous and unbounded, but is for example discrete or strictly positive.

  In Angrist and Pischke (2008)'s view, the structure of the outcome variable is irrelevant, **linear regression** is always appropriate as the population linear projection **provides the best mean-square *linear* approximation to the conditional expectation function** (CEF) $\mathbb{E}[Y_i \mid D_i]$.[15]

  - *Simplest case: binary D, no* X.

    Because D_i takes only the values zero and one, its CEF is necessarily linear in D_i:

    $$\mathbb{E}[Y_i \mid D_i] = \mathbb{E}[Y_i \mid D_i{=}0] + (\mathbb{E}[Y_i \mid D_i{=}1] - \mathbb{E}[Y_i \mid D_i{=}0])\, D_i.$$

    Thus, regardless of the constraints on the support of Y_i, we have $\beta_{\text{OLS}} = \mathbb{E}[Y_i \mid D_i = 1] - \mathbb{E}[Y_i \mid D_i = 0]$, which, absent selection bias, identifies the ATET.

  - *General case: nonbinary D or additional covariates.*

    The CEF $\mathbb{E}[Y_i \mid D_i, \mathrm{X}_i]$ is generally nonlinear for limited Y_i (the saturated-covariate specification is impractical, and Y isn't normal so (Y, D, X) isn't multivariate normal). Linear regression then provides not the CEF itself, but still the best linear approximation to the CEF. As before, the CEF is causal under the CIA (i.e., identifies the conditional causal response function). Linear regression thus identifies the best linear approximation to the causal effect.[16]

[15] The question this raises however, is, in cases of a *highly nonlinear* CEF, what would we really learn from a linear approximation to it? A CEF is actually linear under only two rare circumstances: if (Y_i, D_i) is multivariate normal, or if the model is saturated (i.e., it has a separate parameter for each possible combination of values of the regressors).

[16] However, as the regression vector misses some features of the CEF, it would most likely generate fitted values outside the support of Y. This is for example a well-known problem of the linear probability model. For bounded or discrete outcomes, nonlinear conditional-mean models may fit the CEF better. But if one is interested only in estimating marginal effects (the average changes in CEF), this might be not so consequential.

2.2 The causal graph framework

An alternative to the potential outcomes (PO) framework for addressing causality has gained traction in disciplines other than economics: the structural causal model framework, and more precisely its representation using directed acyclic graphs (DAGs), largely developed by J. Pearl (Pearl, 2009). This section merely introduces it.[17]

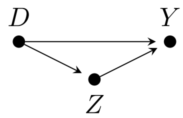
The two frameworks are not opposed: both define causal effects through counterfactual outcomes. While the potential outcomes framework indexes these counterfactual outcomes directly, structural causal models encode how interventions propagate through a system.[18] Both frameworks have benefits. They are therefore complementary perspectives, and it can be useful to frame a causal analysis in the language of each.

2.2.1 Elements of directed acyclic graphs

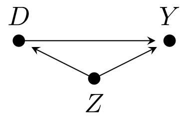
Relationships are encoded using nodes and directed edges. Nodes represent random variables—circles are solid if they are observed and hollow otherwise—while arrows represent possible direct causal relationships. A path is any sequence of adjacent edges. It is *open* if it can generate an association between its endpoints, and *closed* or *blocked* otherwise. Conditioning on variables along the path may open or close it.

Three elementary path structures influence the association between D and Y:

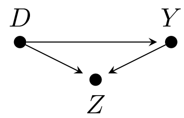
- **mediating path:** D causally affects Y through a mediator Z.
  ↪ *Conditioning on Z blocks the causal path through Z, thereby removing the component of the effect transmitted through it. Not conditioning identifies the total effect of D on Y; conditioning may identify the direct effect.*



- **confounding path:** a confounder Z causally affects both D and Y.
  ↪ *Z creates a non-causal association between D and Y. Conditioning on Z blocks this path, and identifies the total causal effect of D on Y.*[19]



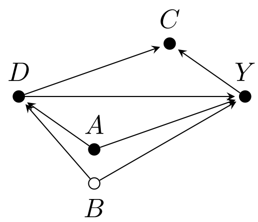
- **colliding path:** a collider Z is causally affected by both D and Y.
  ↪ *The path through Z is blocked by default. Conditioning on Z, or on one of its descendants, opens it and induces an additional non-causal association in our estimation of the relationship between D and Y.*



A **back-door path** from D to Y is any path between them whose first edge is an arrow pointing to D.

Importantly, a causal DAG is a complete encoding of assumptions about direct causal relationships: those assumed to exist represented by arrows, and those assumed to not exist represented by missing arrows. I.e., the absence of an arrow is not a lack of assumption, but a structural exclusion restriction.

For example, the basic DAG on the right encodes:

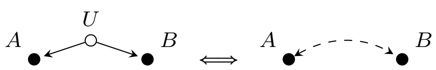


⋆ explicitly, four paths linking D to Y:
  $D \longrightarrow Y$: a direct causal path
  $D \longleftarrow A \longrightarrow Y$: a back-door confounding path, blocked if conditioning on A
  $D \dashleftarrow B \dashrightarrow Y$: a back-door confounding path, open as B is unobserved
  $D \longrightarrow C \longleftarrow Y$: a colliding path
⋆ implicitly, three assumptions of no direct relationships between A, B and C.

[17] For a detailed presentation, see Morgan and Winship (2015, ch. 1.5 & 3), of which this section is a proposed summary.

[18] How directed graphs encode causal states is not detailed here. See Sections 3.4 and 3.6 of Morgan and Winship (2015), or Pearl (2009), for a detailed presentation. Importantly, we also consider only the subset of directed *acyclic* graphs (DAGs), where no directed paths emanating from a causal variable also terminate at the same causal variable. This prohibition of cycles notably rules out simultaneous causality and feedback loops. Section 3.2 of Morgan and Winship (2015) discusses the implications.

[19] If the confounder was unobserved, we could either represent it using the same arrows and an empty node, as per our definitions, or using a curved dashed bidirected edge as in the figure below as a shorthand. Such bidirected edges should not be interpreted as mere correlations between the two variables, they represent an unspecified set of unobserved common causes of the two variables that they connect.



2.2.2 Two main identification strategies: 1. Blocking back-door paths; 2. Instruments

We want to estimate the causal effect of a treatment D on Y. We represent in a causal graph this relationship and all other relationships relevant to its identification. *Given the structure of the causal relationships, which variables must we observe and condition on to identify the causal effect of D on Y?*

- Strategy 1: Blocking back-door paths

  The most common concern with observational data is that an open back-door path creates a non-causal association between D and Y. **The total association between D and Y identifies the total causal effect if there are no open back-door paths.**

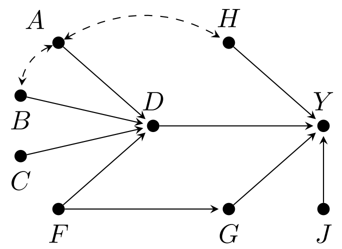
  – In the previous basic DAG:

  If there is no B, conditioning on A closes the back-door path through it and identifies the causal effect. With an unobserved B, an open back-door path remains ⇒ back-door adjustment is unavailable.

  – In the richer DAG on the right, there are three back-door paths:

  $D \longleftarrow A \dashleftarrow\!\!\dashrightarrow H \longrightarrow Y$
  $D \longleftarrow B \dashleftarrow\!\!\dashrightarrow A \dashleftarrow\!\!\dashrightarrow H \longrightarrow Y$
  $D \longleftarrow F \longleftrightarrow G \longrightarrow Y$

  We can block all back-door paths by either:
  ∗ conditioning on H and either F or G;
  ∗ conditioning on A and B,[20] and either F or G.



- Strategy 2: Instruments

  Instead of blocking back-door paths, we can use exogenous variation in an instrument[21] Z to isolate variation in D that is unrelated to the unobserved causes of Y. In the graph above, C is a candidate instrument for D; so is F after conditioning on G.
  ⚠ *The graph only represents exclusion and independence restrictions; the exact causal estimand identified additionally depends on assumptions such as functional form, effect homogeneity, or monotonicity.*

→ To identify the causal effect of D on Y, we reach the same general conclusion as with the PO framework:

> In observational studies,
> – An open back-door path generally makes the observed association between D and Y differ from the causal effect.
> – *Back-door criterion:* Blocking all back-doors by conditioning on a relevant set of variables identifies the total causal effect.
> – Because we can rarely be certain that a credible adjustment set is available, we typically turn to alternative identification strategies, that rely on other assumptions.

2.3 Comparative strengths and weaknesses of the PO and DAG approaches

The potential outcomes framework being the most popular in econometrics, we ask what the DAG approach adds, and the ways in which it differs that are most relevant for work in econometrics.[22]

- **Role of experiments and manipulability**

  While the PO literature elevates randomized experiments as gold standard, the graphical literature treats randomization as one possible intervention and graph structure rather than as its main organizing principle. Related to this is the notion of manipulability:

[20] Conditioning only on A would not suffice. As A is a collider along the path between B and H, conditioning only on A would create dependence between B and H, and so wouldn't eliminate the noncausal association between D and Y.

[21] Instruments are formally introduced in the next section. In short, a variable Z is a valid instrument for D if it does not have an effect on Y except through its effect on D. We can then estimate the effect of D on Y by taking the ratio of the relationships $Z \leftrightarrow Y$ and $Z \leftrightarrow D$.

[22] Imbens (2020) proposes such a review, of which the majority of this section is a proposed summary.

- The PO framework usually defines potential outcomes with reference to a well-specified intervention or manipulation. This conceptual framework of an intervention has the benefit of clarifying the treatment and its policy relevance[23]—and is arguably thereby justified for economics where *"policy relevance is a key goal"* (Imbens, 2020).[24] On the other hand, the implicit criterion of manipulability may be potentially restrictive and unnecessary. The DAG literature instead does not deny causal character to nonmanipulable variables.

- **Parts of the causal analysis addressed**

  Consider three parts of a causal analysis: (1) pre-identification: developing a causal model; (2) identification: establishing whether a causal estimand is determined by the observed-data distribution; and (3) post-identification: estimation and statistical inference from a sample.

  - Neither framework helps much with (1) (postulating a causal model of how the world works).
  - The graphical literature is particularly developed for (2), and rather silent on (3). We note that DAGs encode *nonparametric* causal relationships; no assumptions are made about functional forms or distributions. All interactions between the different variables affecting another are also permitted (an arrow into Y signifies inclusion in the arbitrary structural function $f_Y(D, X, \ldots)$).
  - The PO-based literature often addresses (2) and (3) jointly and focuses on research design, estimation, and inference—for example, weak instruments, matching, robust inference...

- **Representation of identifying assumptions and identification strategies**

  👍 Assumptions concerning the existence or absence of causal relationships are explicit in a DAG and often clearer than their algebraic or counterfactual expressions. In IV settings, for example, missing arrows visually represent exclusion and independence restrictions.

  👎 Other assumptions are not easily represented using an ordinary DAG, particularly shape restrictions such as monotonicity, continuity, or convexity. Yet these play an important role in identification strategies such as IV and RD. In RD, for example, identification relies on a discontinuity in treatment assignment and continuity of the potential-outcome regressions, which a DAG does not make especially transparent; see Steiner et al. (2017).

  👍 The graphical literature provides systematic machinery for deriving identification and selecting adjustment variables, including the back-door and front-door criteria and do-calculus.[25] It also shows why 'controlling for all other causes of Y' can be misleading. In the previous graph, for example, we found two valid adjustment sets: either H or the combination of A and B—in each case together with either F or G.

  👎 Ordinary DAGs do not display treatment-effect heterogeneity or latent compliance types such as compliers, never-takers, and always-takers, directly (although other structural causal models can). The PO framework often represents estimands such as LATEs more transparently.

Imbens (2020) suggests further practical reasons for the limited adoption of DAGs in economics:

- The graphical literature has provided relatively few realistic empirical examples in which DAG methods clearly improve upon established econometric approaches. By contrast, the PO framework gained influence through applications demonstrating the practical value of its methods.
- Standard DAGs are acyclic and therefore do not directly represent feedback loops, simultaneity, or equilibrium behavior, which are central to many economic models and accommodated in the PO framework.

[23] For example, consider a study comparing hiring outcomes when the racial-sounding of applicant names is changed. The conceptual framework of a manipulation helps clarify that the study cannot estimate the causal effect of race itself. What it captures is the well-defined causal effect of manipulation of the perception of race.

[24] Imbens (2020) writes: *"It is also not obvious to me why we would care [...] if the effect is not tied to an intervention we can envision."* and notes that much of the empirical work in economics focuses on questions about manipulations, e.g., *"What would happen to my headache if I take an aspirin?"* or *"How effective is a given treatment in preventing a disease?"*

[25] The front-door criterion is not developed here as it relies on exclusion restrictions that seem unrealistic in many social science applications. As Imbens (2020) points out, such difficulty in specifying credible models in economics *"was a big part of the motivation for the so-called credibility revolution, with its focus on natural experiments."*

3 Design stage: canonical identification strategies

Hierarchy of common identification methods A contestable hierarchy of the most common identification methods in the 'randomista' toolkit, based on their capacity to mimic random assignment, is as follows:

0. Randomized experiments (RCT) and natural experiments where treatment assignment is as-if random
1. Instrumental variables (IV) and regression discontinuity (RD)
   An instrument or discontinuity generates plausibly exogenous variation in treatment.
2. Difference-in-differences (DiD), event studies, and synthetic control methods (SCM)
   Repeated observations are used to construct a comparison trajectory that represents how the treated units would have evolved absent their exposure to treatment at a given event time.
3. Matching
   Strategies based solely on matching are considered much less credible than strategies based on some exogenous variation in treatment. Indeed, they require that all confounders are correctly accounted for (the assumed assignment mechanism is by selection on observables) and overlap. Matching should therefore generally complement, rather than replace, a credible natural- or quasi-experimental design. It is presented in tht context in Section 4.

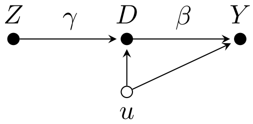
Each method is presented according to the following structure: (i) the design, notably the treatment-assignment mechanism; (ii) the identifying assumptions; (iii) the estimand, or treatment effect of interest; (iv) the estimator; and (v) key best practices, strengths, and weaknesses.

Importantly, we show for each method how observed outcomes Y_i, under its identifying assumptions, recover a causal estimand defined in terms of potential outcomes Y_i^0, Y_i^1. This connection is crucial: it shows why a particular comparison constructed from observed data can be interpreted as a causal effect.

For simplification purposes, the methods are mostly presented without the inclusion of exogenous controls X_i, but they could—and arguably should—be generalized to conditioning on such covariates.

3.1 Instrumental variables

Design $D_i \in \{0,1\}$ is an endogenous treatment and $Z_i \in \{0,1\}$ is a binary instrument that is a random source of variation in D_i by changing the probability of treatment. We can use the instrument to isolate variation in D that identifies the average treatment effect among compliers.[26]



$$D_i = \delta + \gamma Z_i + v_i$$
$$Y_i = \alpha + \beta D_i + u_i, \qquad \text{cov}[D_i, u_i] \neq 0$$

Potential treatments
Let $D_i^z \in \{0,1\}$ denote individual i's potential treatment status under $Z = z$. The pair $\left(D_i^0, D_i^1\right)$ defines four latent compliance types—compliers, always-takers, never-takers, and defiers—which the researcher does not observe. For a given individual i, $Z_i = 0$ induces the potential treatment status D_i^0, realized as 0 if i complies and 1 otherwise; $Z_i = 1$ generates the potential treatment status D_i^1, realized as 1 if i complies and 0 otherwise.

	D_i^0	D_i^1
compliers	0	1
always-takers	1	1
never-takers	0	0
defiers	1	0

Identifying assumptions

(A1) *Exclusion restriction* (no direct effect of Z on Y): Z affects Y only through D. → The effect of Z on Y—aka the intention to treat (ITT)—is zero for never-takers and always-takers, whose treatment status is unchanged by Z.

(A2) *Independence* (of Z from the potential outcomes and potential treatments), or "as-if random assignment": $Z_i \perp\!\!\!\perp (Y_i^0, Y_i^1, D_i^0, D_i^1)$. I.e., there is no confounder that affects both the instrument and the outcome, nor one that affects both the instrument and treatment take-up.

(A3) *Relevance* (of Z): $\text{cov}[Z_i, D_i] \neq 0$, or equivalently, $\gamma \neq 0$.

(A4) *Monotonicity* (of the effect of Z on D): $\forall i,\ D_i^1 \geq D_i^0$. Z does not discourage treatment for any individual (no defiers). → Compliers are the only individuals whose treatment status is altered by the instrument.

Estimand $$\beta_{\text{IV}} := \frac{\text{cov}[Y_i, Z_i]}{\text{cov}[D_i, Z_i]} = \dots = \frac{\mathbb{E}[Y_i \mid Z_i{=}1] - \mathbb{E}[Y_i \mid Z_i{=}0]}{\mathbb{E}[D_i \mid Z_i{=}1] - \mathbb{E}[D_i \mid Z_i{=}0]} = \dots = \underbrace{\mathbb{E}[Y_i^1 - Y_i^0 \mid D_i^0{=}0, D_i^1{=}1]}_{\textbf{LATE for compliers}}$$

Estimator A natural choice of estimator is the sample analog called *Wald estimator* $\hat{\beta}_{\text{W}} = \frac{\widehat{\text{cov}}[Y_i, Z_i]}{\widehat{\text{cov}}[D_i, Z_i]}$. $\hat{\beta}_{\text{W}}$ turns out to be numerically equivalent to the two-stage least squares (2SLS) estimator $\hat{\beta}_{\text{2SLS}}$ obtained through the two-step process[27]

$$\begin{aligned} \text{First stage:} \quad & D_i = \delta + \gamma Z_i + v_i, \qquad && \Rightarrow \quad \widehat{D}_i = \widehat{\delta} + \widehat{\gamma} Z_i, \\ \text{Second stage:} \quad & Y_i = \widetilde{\alpha} + \widetilde{\beta}\, \widehat{D}_i + e_i. && \end{aligned}$$

Note: The reduced form of a model is that where the endogenous variables are expressed as functions of the exogenous variables. In the IV setting, the regression of Y on Z is therefore often called the reduced form, and estimates the intention-to-treat effect of the instrument on the outcome, whereas the Wald/IV estimand identifies the LATE among compliers.

[26] For more general treatment variables, we will need more advanced instruments. Ex: to identify *several* treatment variables, we will need at least as many instruments; to identify a *continuous* treatment, we can't use a binary instrument.

[27] While the coefficients are equal, the SEs of the second stage would not give the correct SEs, as we need to adjust for the two stages of estimation. We must account for the estimation uncertainty from the first-stage (the first-stage is based on a sample, not the population, making $\widehat{D_i}$ a random variable, instead of the usual fixed variable). Most 2SLS packages do the adjustment automatically—otherwise one can simply bootstrap the SEs a posteriori.

Best practices

- Support the relevance assumption by reporting a large F-statistic for the first stage (conventional rule of thumb: $F > 10$). The higher F, the stronger the instrument.
- As in any observational study, consider adjusting for important pretreatment predictors to improve precision and reduce model sensitivity, making sure to include the same covariates in both stages.
- Different valid instruments select different sets of compliers, leading to different estimands. Consider the specific group of compliers selected and make sure the instrument is relevant w.r.t. the policy of interest. Then count and characterize these compliers to get more out of the LATE (see below).
- For nonlinear models, the properties of 2SLS do not necessarily hold, so one may want to consider alternative estimation strategies, such as the *control function method:*
  - Estimation is in two stages: (i) same first stage, extract the residuals $\hat{v}$; (ii) OLS regression of Y on $(D, \hat{v})$.
  - This comes at the cost of additional functional-form and distributional restrictions.
  - In the standard linear model, the control function gives the same coefficient as 2SLS.

Strengths and weaknesses

- Compelling identification strategy
- IV estimates commonly have larger standard errors than the corresponding OLS estimates, and their precision further decreases with weak instruments. *Note however that they do not estimate the same causal parameter, so this is not an efficiency comparison.*
- 2SLS has finite-sample bias, which stems from the randomness in estimates of $\widehat{D}_i$ and increases with the weakness and number of instruments. Weak instruments can even render $\hat{\beta}_{\text{IV}}$ more biased than $\hat{\beta}_{\text{OLS}}$.[28] See Andrews et al. (2019).
- In many settings (e.g., models non-linear in D, non-saturated models with covariates), 2SLS can be very biased. New paper: `https://a-torgovitsky.github.io/tsls-weights.pdf`
- One can also use IVs to address the attenuation bias that may result from measurement error in D. Ex: Krueger and Lindahl (2001) uses an IV to address attenuation bias in cross-country estimates of the returns to education.

Counting and characterizing compliers to get more out of the LATE Compliers ($D_i^1 > D_i^0$) need not be representative of the population, due to selective uptake. While we cannot identify individual compliers in the data, we can (i) estimate the share of the complier group, and (ii) characterize them in terms of their observed covariates.

- *Counting compliers:* We can measure the share of the compliers among the whole population and among the treated (Angrist and Pischke, 2008, 4.4.4):
  - In the unconditional identification case:

$$\Pr(D_i^1 > D_i^0) = ... = \mathbb{E}[D_i \mid Z_i{=}1] - \mathbb{E}[D_i \mid Z_i{=}0] \quad \text{, which is the raw first stage}$$

$$\Pr(D_i^1 > D_i^0 \mid D_i{=}1) = \ldots = \frac{\Pr(Z_i{=}1) \times (\mathbb{E}[D_i \mid Z_i{=}1] - \mathbb{E}[D_i \mid Z_i{=}0])}{\Pr(D_i{=}1)} = \frac{\text{share}_{Z_i=1} \times \text{1st stage}}{\text{share}_{\text{treated}}}$$

  - When instrument validity holds only conditional on X_i:

$$\Pr(D_i^1 > D_i^0) = ... = \mathbb{E}_{\mathrm{X}}\big[\mathbb{E}[D_i \mid Z_i{=}1, \mathrm{X}_i{=}\mathrm{x}] - \mathbb{E}[D_i \mid Z_i{=}0, \mathrm{X}_i{=}\mathrm{x}]\big]$$

$$\Pr(D_i^1 > D_i^0 \mid D_i{=}1) = \ldots = \frac{\mathbb{E}_{\mathrm{X}}\big[\Pr(Z_i{=}1 \mid \mathrm{X}_i) \times (\mathbb{E}[D_i \mid Z_i{=}1, \mathrm{X}_i{=}\mathrm{x}] - \mathbb{E}[D_i \mid Z_i{=}0, \mathrm{X}_i{=}\mathrm{x}])\big]}{\Pr(D_i{=}1)}$$

- *Characterizing compliers:* We can describe the distribution of covariates X for compliers.

[28] Note that these are different biases: endogeneity bias with the OLS estimator and sample bias with the IV estimator.

- For a binary characteristic, we can calculate relative likelihoods (Angrist and Pischke, 2008, 4.4.4). For example, the likelihood that a complier has $X_i = 1$, relative to the corresponding probability in the overall population, is:

$$\frac{\Pr(X_i{=}1 \mid D_i^1 > D_i^0)}{\Pr(X_i{=}1)} = \ldots = \frac{\mathbb{E}[D_i \mid Z_i{=}1, X_i{=}1] - \mathbb{E}[D_i \mid Z_i{=}0, X_i{=}1]}{\mathbb{E}[D_i \mid Z_i{=}1] - \mathbb{E}[D_i \mid Z_i{=}0]} = \frac{\text{1st stage} \mid X_i{=}1}{\text{1st stage}}$$

- For general covariates, we can calculate the mean—or other features of the distribution—of the covariate for compliers using Abadie (2003)'s kappa-weighting scheme. Suppose the identifying assumptions hold conditional on X_i, and suppose overlap, i.e., $0 < \Pr(Z_i{=}1 \mid \mathrm{X}_i) < 1$. For any function $g(Y_i, D_i, \mathrm{X}_i)$ with finite expectation, we have

$$\mathbb{E}[g(Y_i, D_i, \mathrm{X}_i) \mid D_i^1 > D_i^0] = \frac{\mathbb{E}[\kappa_i\, g(Y_i, D_i, \mathrm{X}_i)]}{\mathbb{E}[\kappa_i]},$$

with the weighting function $\kappa_i = 1 - \frac{D_i(1-Z_i)}{1-P(Z_i=1|\mathrm{X}_i)} - \frac{(1-D_i)Z_i}{P(Z_i=1|\mathrm{X}_i)}$.

- Applications: See Almond and Doyle (2011); Kowalski (2021).

Shift-share instruments Shift-share variables combine a common set of sector-level shifts with heterogeneous exposure shares: $Z_i = \sum_k s_{ik} g_k$, where s_{ik} measures unit i's initial exposure to sector k, and g_k is the corresponding sector-level shift. They are widely used in labor, trade, macroeconomics, public economics, and finance to study settings in which units are differentially exposed to a common set of shocks (Borusyak et al., 2025).

Identification may rely either on the shifts being plausibly exogenous, conditional on the shares, or on the exposure shares being plausibly exogenous, conditional on the shifts. The shifts and shares are therefore components of the instrument.

Examples:

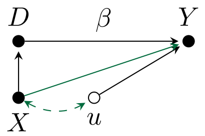
- Autor et al. 2013 study how the surge in Chinese imports in the 1990s and 2000s affected U.S. local labor markets.
  - Their endogenous shift-share explanatory variable measures the U.S. regional exposure to the Chinese import competition shock, i.e., the extent to which workers were employed in industries that saw growing competition with China: $X_i = \sum_k s_{ik} \Delta M_k^{US}$ where s_{ik} is the initial share of workers in region i employed in industry k, and ΔM_k^{US} is the subsequent growth of U.S. imports from China in that industry. X is hence the average of national industry-level shifts in US imports from China, weighted by the regional shares of employment across industries.
  - X is potentially endogenous, as U.S. imports from China may respond to U.S.-specific productivity or demand shocks. The authors instrument this exposure using: $Z_i = \sum_k s_{ik} \Delta M_k^{O}$, where ΔM_k^{O} is the growth of Chinese exports to other high-income countries in industry k. The instrument therefore uses a similar shift-share structure, with the same initial local U.S. exposure shares but replaces U.S. import growth with Chinese export growth in other countries, intended to capture supply-driven changes in China's export capacity rather than U.S.-specific shocks.

3.2 Regression discontinuity

Sharp RD (discontinuity in treatment assignment at X=c)

Design Treatment D_i is not randomly assigned: it is deterministically but *discontinuously* assigned according to whether a continuous pretreatment *running variable* X_i crosses a cutoff c: $D_i = \mathbb{1}\{X_i \geq c\}$. Assuming that the conditional expectations of both potential outcomes vary continuously with X_i at the cutoff, a discontinuity in $\mathbb{E}[Y_i | X_i]$ at c can be attributed to the causal effect of D_i, i.e., identifies the average treatment effect at the cutoff.[29]

The data generating process (DGP) is represented in reduced-form and in the causal graph below.[30] $f(X_i, \phi_{D_i})$ captures the component of the outcome that varies continuously with X_i, βD_i allows $\mathbb{E}[Y_i | X_i]$ to jump at the cutoff, and the green pathways remain smooth at the cutoff and therefore cannot generate a jump in Y_i.



$$Y_i = \alpha + \beta D_i + f(X_i, \phi_{D_i}) + u_i$$

⚠ *There is no conditional overlap. The counterfactual cannot constructed by comparing treated and untreated observations at the same X_i—those do not exist; treated and untreated have different X_i by definition. The comparison is instead between the limits of the outcome regression as X_i approaches c, obtained by extrapolating locally to c from either side. This implies that we may not be that agnostic about functional form... But by using only observations in a small neighborhood around c, we reduce dependence on functional-form assumptions away from c.*

Identifying assumptions

(A1) *Local continuity:* $\mathbb{E}[Y_i^1 | X_i{=}x]$ and $\mathbb{E}[Y_i^0 | X_i{=}x]$ are continuous in x at c. That is, the effects of other determinants of Y_i don't jump at c. $\implies$ the expected potential outcomes of units immediately below the cutoff (who don't get treated) are valid counterfactuals for units immediately above it (who do).

(A2) *Relevance and local support:* Treatment jumps at c, and observations exist close to c on both sides.

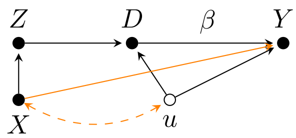
Estimand $\beta_{\text{RD}} = \lim_{x \to c^+} \mathbb{E}[Y_i \mid X_i{=}x] - \lim_{x \to c^-} \mathbb{E}[Y_i \mid X_i{=}x] \;=\; \ldots \;=\; \underbrace{\mathbb{E}[Y_i^1 - Y_i^0 \mid X_i{=}c]}_{\textbf{LATE at the cutoff}}$

Estimator We can estimate the discontinuity in $\mathbb{E}[Y_i | X_i]$ at the cutoff using separate smooth regression functions on either side (to capture different trends for $\mathbb{E}[Y_i^1 | X_i]$ and $\mathbb{E}[Y_i^0 | X_i]$), with the regression:

$$Y_i = \alpha + \beta D_i + f(X_i, \phi_l) + f(X_i, \phi_r) D_i + e_i$$

Fuzzy RD (discontinuity in Pr(treatment assignment) at X=c)

Design At the cutoff c, the running variable X_i generates a discontinuity not in treatment assignment itself, but in the probability of treatment: $\Pr(D_i{=}1 \mid X_i)$. The cutoff indicator $Z_i \coloneqq \mathbb{1}\{X_i \geq c\}$ therefore provides an instrument for the treatment status D_i. The DGP is represented in the causal graph below: the orange pathways remain smooth at the cutoff and therefore cannot generate a jump in Y_i.



[29] ⚠ In RD, the observed conditional mean $\mathbb{E}[Y_i \mid X_i{=}x]$ is not assumed smooth at c; its discontinuity is precisely what identifies the treatment effect. The assumption is that the two potential-outcome regression functions are separately continuous: $\mathbb{E}[Y_i^0 | X_i{=}x]$ and $\mathbb{E}[Y_i^1 | X_i{=}x]$ at $x = c$.

[30] The causal graph is taken from Steiner et al. (2017).

Identifying assumptions Let Y_i^z and D_i^z denote the potential outcome and treatment status, respectively, under $Z_i = z$.

(A1) *Exclusion restriction:* crossing the cutoff affects Y_i only through its effect on treatment status D_i. Equivalently, conditional on treatment status, Z_i does not itself generate a discontinuity in the potential-outcome regression functions at c.

(A2) *Local continuity:* the conditional expectations of the relevant potential outcomes and potential treatments, namely, for $z \in \{0,1\}$, $\mathbb{E}[Y_i^z \,|\, X_i{=}x]$ and $\mathbb{E}[D_i^z \,|\, X_i{=}x]$, are continuous in x at c.[31]

(A3) *Relevance and local support:* the probability of treatment jumps at the cutoff: $\lim_{x\to c^+} \mathbb{E}[D_i \,|\, X_i{=}x] \neq \lim_{x\to c^-} \mathbb{E}[D_i \,|\, X_i{=}x]$, and observations exist arbitrarily close to c on both sides.

(A4) *Local monotonicity:* $D_i^1 \geq D_i^0$ for units at the cutoff: crossing the cutoff does not discourage treatment for any unit (no defiers).

Estimand $$\beta_{\text{RD}} := \frac{\lim\limits_{x\to c^+} \mathbb{E}[Y_i \mid X_i{=}x] - \lim\limits_{x\to c^-} \mathbb{E}[Y_i \mid X_i{=}x]}{\lim\limits_{x\to c^+} \mathbb{E}[D_i \mid X_i{=}x] - \lim\limits_{x\to c^-} \mathbb{E}[D_i \mid X_i{=}x]} = \ldots = \underbrace{\mathbb{E}\left[Y_i^1 - Y_i^0 \mid X_i{=}c,\ D_i^1{>}D_i^0\right]}_{\textbf{LATE for compliers at the cutoff}}$$

Estimator Fuzzy RD naturally lends itself to a 2SLS estimation strategy. The 2SLS estimator obtained through the two-step process below is numerically equivalent to the ratio of the reduced-form jump in Y_i to the first-stage jump in D_i.

$$\begin{aligned}
\text{First stage:} \quad & D_i = \delta + \gamma Z_i + f(X_i, \phi_l) + Z_i f(X_i, \phi_r) + v_i, \quad \Rightarrow \widehat{D}_i = \widehat{\delta} + \widehat{\gamma} Z_i + f(X_i, \widehat{\phi}_l) + Z_i f(X_i, \widehat{\phi}_r), \\
\text{Second stage:} \quad & Y_i = \widetilde{\alpha} + \widetilde{\beta}\, \widehat{D}_i + f(X_i, \widetilde{\phi}_l) + Z_i f(X_i, \widetilde{\phi}_r) + e_i.
\end{aligned}$$

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Best practices

- Choice of $f(\cdot, \phi)$: this function is unknown. To avoid misspecification of the functional form of the DGP, the regression functions at the boundary point are generally estimated using flexible functional forms, such as local low-order polynomial regressions:
  - linear: $f(X_i, \phi) = \phi(X_i - c)$,
  - quadratic: $f(X_i, \phi) = \phi_a(X_i - c) + \phi_b(X_i - c)^2$,

  with $c - h \leq X_i \leq c + h$.[32]

  Results should be examined for sensitivity to bandwidth and low polynomial order. ⚠ Global or high-order polynomials can lead to overfitting and introduce bias (Gelman and Imbens, 2019).
- The densities of the running variable and of predetermined covariates should not jump at the cutoff. Check whether they do: such discontinuities may indicate manipulation or sorting that undermines the continuity assumption.
- In fuzzy RDs, examine and report the first-stage discontinuity in treatment probability. A small first-stage jump produces a weak and imprecise fuzzy-RD estimate.
- As in any observational study, consider adjusting for important pretreatment predictors to improve precision and reduce model sensitivity.

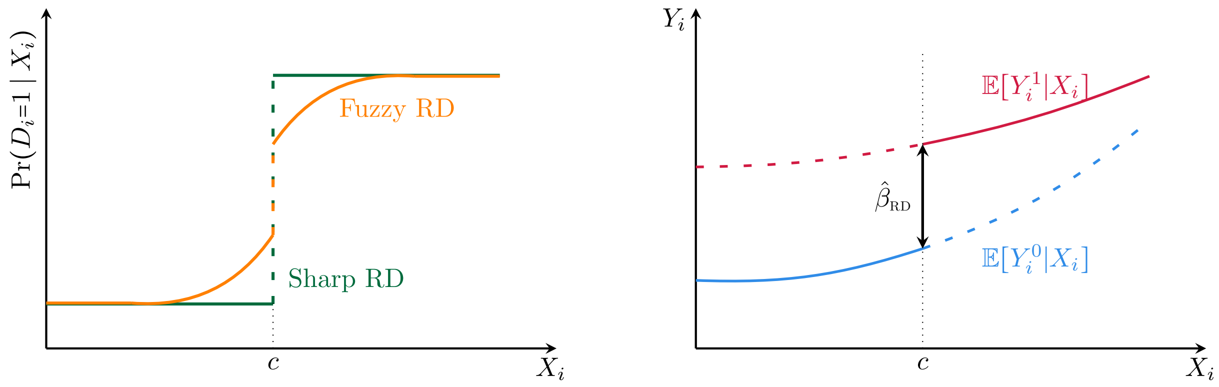
[31] In fuzzy RD, crossing the cutoff changes the instrument Z, not treatment D deterministically. Immediately below and above c, we observe respectively the potential outcomes under the two instrument states: $Y_i^z := Y_i(D_i(z))$, assuming exclusion. Using only $\mathbb{E}[Y_i^d | X_i{=}x]$, $d \in 0, 1$, is generally insufficient, because the observed outcome under $Z = z$ also depends on who takes treatment under that instrument state. In contrast, in sharp RD, $D_i^z = z$, so under exclusion we have $Y_i^z = Y_i^{d=z}$.

[32] A larger bandwidth h increases precision but also bias. One may choose the optimal h by estimating the model's predictive accuracy for different values of h, for example using leave-one-out cross-validation: iteratively for each observation i, fit the model using only the observations $X_i{-}h \leq \mathbf{X} < X_i < c$ when $X_i < c$, and only the observations $c < X_i < \mathbf{X} \leq X_i{+}h$ when $X_i \geq c$.

Strengths and weaknesses

- 👍 RD designs essentially approximate a local randomized experiment around the cutoff, and can thereby credibly identify a local causal effect under relatively weak continuity assumptions (although fuzzy RDs require additional exclusion and monotonicity assumptions to interpret the estimand as a LATE).
- 👍 They exploit jumps in the probability of treatment as we move along some X. They have much potential in economic applications, as geographic boundaries or administrative or organizational rules (e.g., program eligibility thresholds) often create usable discontinuities.
- 👎 They risk being underpowered because identification relies primarily on observations close to the cutoff.
- 👎 Fuzzy RDs may be especially underpowered when the discontinuity in treatment probability is small.
- 👎 RD estimates are highly local, so their external validity may be limited.

Extensions See also dynamic RD designs (Cellini et al., 2010), and *differences in discontinuities* (RDs with DiD).



(a) Treatment assignment (sharp & fuzzy) (b) Treatment effect (sharp)

Figure 1: RD treatment assignments and effects

3.3 DiD, DiDiD, event studies, and synthetic control method

Repeated observations help adjust for unobserved confounding

- In linear fixed-effects models, repeated observations along some dimension allow us to absorb unobserved confounders whose effects are constant across that dimension and enter the model additively.

  Say we observe each individual i repeatedly over periods $t = 1, \dots, T$. We can include several sets of fixed effects to absorb several kinds of additive confounding:

  - with individual fixed effects, β is estimated from variation within individuals over time;[a]
  - with period fixed effects, β is estimated from variation across individuals within periods;
  - with both individual and period fixed effects (*two-way fixed effects*), β is estimated from the variation remaining after removing both (i) individual-specific levels and (ii) shocks common to all individuals in each period.[b]

  β is a weighted regression average of multiple comparisons. Under additional identifying assumptions, it may be interpreted as a weighted aggregation of the corresponding treatment effects.

- Other methods use repeated observations differently. In particular, the synthetic control method (SCM) does not remove confounding through fixed effects or demeaning, but uses pre-treatment matching.

[a]Including individual fixed effects in a linear model, $Y_{it} = \alpha_i + \beta X_{it} + u_{it}$, is equivalent to manually demeaning the variables: $Y_{it} - \bar{Y}_i = \beta(X_{it} - \bar{X}_i) + (u_{it} - \bar{u}_i)$. Transforming each variable into its deviation from its individual mean removes individual-specific additive components, i.e., average differences between individuals or the *between-individual* variation, leaving the *within-individual* variation. Similarly, period fixed effects remove changes common to all individuals.

[b]With individual and period fixed effects, each variable is considered relative to both its individual-specific component and its period-specific component. These effects enter additively; this is not equivalent to including individual-by-period fixed effects (which would isolate variation within individual-year, which would be obtained with individual-by-year fixed effects). Here, each of the 'relative to' stands alone.

We consider settings in which units are observed before and after treatment occurs. Estimating a causal effect requires constructing the counterfactual trajectory of the treated units in the absence of treatment. The approaches below construct this trajectory in different ways:

- **DiD and DiDiD** Some units remain untreated or are treated later. Under no anticipation and parallel trends in untreated potential outcomes, the *between-group* comparison of *within-group* changes identifies treatment effects for the treated (i.e., we use changes among the comparison units to remove counterfactual changes among the treated). DiDiD adds a third comparison dimension to remove an additional source of differential change.
- **Event studies** Units are treated at different times and may all eventually be treated. Event studies extend DiD by estimating dynamic post-treatment effects and pre-treatment coefficients used to assess parallel trends and anticipation. They rely on the same core identifying assumptions as DiD, albeit applied across periods and treatment cohorts, and use both *within* and *between* variation.
- **SCM** One or a few units are treated. When parallel trends between individual units are implausible, unaffected units from a *donor pool* are combined to reproduce the treated unit's pre-treatment trajectory. The synthetic unit's post-treatment trajectory then estimates the treated unit's counterfactual trajectory.

DiD

Design Treatment assignment is a function of two dimensions: group (treatment/control) and most commonly time (pre/post exposure at τ).[33] We define the associated binary variables $G_i := \mathbb{1}\{i \in \text{treatment group}\}$ and $P_t := \mathbb{1}\{t \in \text{post period}\} = \mathbb{1}\{t \geq \tau\}$, such that treatment status is $D_i = G_i P_t$. In the simple two-groups two-periods case below, $t = 0$ and $t = 1$ denote the pre- and post-treatment periods.

- In a before/after comparison within the treatment group, the observed change in mean outcomes could reflect other changes occurring over the same period (time effects), rather than treatment alone.
- In a treatment/control comparison within the post period, the observed difference in mean outcomes could reflect permanent differences between the groups, rather than treatment alone.
- Instead, the DiD estimand differences out both biases by comparing the *change over time in* mean outcomes in the treatment group to the *change over time in* mean outcomes in the control group.

Identifying assumptions

(A1) *Parallel counterfactual trends across groups:* in the absence of treatment, both groups would have experienced the same *change in untreated potential outcomes*: $\mathbb{E}\big[Y_{i1}^0 - Y_{i0} \mid G_i{=}1\big] = \mathbb{E}\big[Y_{i1}^0 - Y_{i0} \mid G_i{=}0\big]$. This excludes selection into treatment based on differential untreated outcome trends.

(A2) *No anticipation:* treatment at τ does not affect outcomes before τ, so $Y_{it} = Y_{it}^0$ for $t < \tau$.

(A3) *Stable group compositions over time,* unless compositional changes are explicitly adjusted for.

Estimand

$$\beta_{\text{DiD}} := \Big(\mathbb{E}_i\big[Y_{i1} \mid G_i{=}1\big] - \mathbb{E}_i\big[Y_{i0} \mid G_i{=}1\big]\Big) - \Big(\mathbb{E}_i\big[Y_{i1} \mid G_i{=}0\big] - \mathbb{E}_i\big[Y_{i0} \mid G_i{=}0\big]\Big) = \ \dots \ = \underbrace{\mathbb{E}_i\left[Y_{i1}^1 - Y_{i1}^0 \mid G_i{=}1\right]}_{\textbf{ATET in the post period}}$$

Estimator The natural estimator of the population β_{DiD} is its sample analog. It is actually equivalent to the OLS coefficient on $G_i P_t$ in the saturated regression

$$Y_{it} = \alpha + \beta_G G_i + \beta_P P_t + \beta G_i P_t + e_{it}.$$

In this two-groups two-periods design, β_{DiD} is also equal to the interaction coefficient in a two-way fixed-effects regression with group and period fixed effects: $Y_{it} = \lambda_{G_i} + \lambda_t + \beta G_i P_t + u_{it}$. With panel data, group fixed effects λ_{G_i} can be replaced by individual fixed effects α_i.

DiDiD

Design Treatment exposure varies across two cross-sectional dimensions and time. Let G_i indicate membership in the treatment group, S_i indicate membership in the subgroup exposed or eligible for treatment, and P_t indicate the post-treatment period. In the canonical $2 \times 2 \times 2$ design, treatment status is $D_i = G_i S_i P_t$.

Identifying assumptions

(A1) *Parallel counterfactual trends in subgroup differences*: in the absence of treatment, the change in the $S_i = 1$ versus $S_i = 0$ outcome gap would have been the same in the treatment and control groups:

$$\begin{aligned} &\mathbb{E}\left[Y_{i1}^0 - Y_{i0}^0 \mid G_i = 1, S_i = 1\right] - \mathbb{E}\left[Y_{i1}^0 - Y_{i0}^0 \mid G_i = 1, S_i = 0\right] \\ &= \mathbb{E}\left[Y_{i1}^0 - Y_{i0}^0 \mid G_i = 0, S_i = 1\right] - \mathbb{E}\left[Y_{i1}^0 - Y_{i0}^0 \mid G_i = 0, S_i = 0\right]. \end{aligned}$$

(A2) *No anticipation:* treatment at τ does not affect outcomes before τ, so $Y_{it} = Y_{it}^0$ for $t < \tau$.

(A3) *Stable composition within each (G, S) cell over time,* unless compositional changes are adjusted for.

[33] In the canonical DiD setting, the second dimension is time, but it need not be. The same double-difference structure can sometimes be defined over another dimension, such as cohort (e.g., year of birth). Its causal interpretation then requires assumptions appropriate to that design.

Estimand Define $\mu_{gsp} := \mathbb{E}[Y_{it} \mid G_i{=}g, S_i{=}s, P_t{=}p]$. The population triple-difference contrast is

$$\beta_{\text{DIDID}} := \Big[(\mu_{111} - \mu_{110}) - (\mu_{011} - \mu_{010})\Big] - \Big[(\mu_{101} - \mu_{100}) - (\mu_{001} - \mu_{000})\Big] = \ldots = \underbrace{\mathbb{E}\big[Y^1_{i1} - Y^0_{i1} \mid G_i{=}1, S_i{=}1\big]}_{\textbf{ATET in the post period}}.$$

Estimator The natural estimator of the population β_{DIDID} is its sample analog. It is actually equivalent to the OLS coefficient on $G_iS_iP_t$ in the saturated regression

$$Y_{it} = \alpha + \beta_G G_i + \beta_S S_i + \beta_P P_t + \beta_{GS} G_i S_i + \beta_{GP} G_i P_t + \beta_{SP} S_i P_t + \beta_{\text{DIDID}} G_i S_i P_t + e_{it}.$$

Event studies (dynamic DiD with staggered treatment timing)

Design Units are treated at different times and may all eventually be treated. Let τ_i denote the period in which unit i is first treated and define $D_{it} := \mathbb{1}\{t \geq \tau_i\}$, with $k := t - \tau_i$ the time measured relative to the event or *event time*. Event studies extend DiD by estimating treatment effects separately across event times, using not-yet-treated or never-treated units as the comparison group.

Identifying assumptions

(A1) *Parallel counterfactual trends across treatment cohorts:* in the absence of treatment, each treated cohort and the untreated comparison units would have experienced the same change in untreated potential outcomes. This excludes selection into treatment timing based on differential untreated outcome trends.

(A2) *No anticipation:* treatment at τ_i does not affect outcomes before τ_i, so $Y_{it} = Y^0_{it}$ for $t < \tau_i$.

(A3) *Overlap and stable treatment-cohort compositions over time:* for each treatment cohort and event time of interest, an appropriate untreated comparison group is observed, and the composition of the observed sample does not change differentially around treatment in a way related to potential outcomes.

Estimand An event study generally identifies a sequence of treatment effects.

For the cohort with $\tau_i = g$, with $g - 1$ as the baseline period, the effect k periods after treatment is estimated using a DiD comparison between cohort g and units that remain untreated over the relevant comparison period:

$$\text{ATE}_{gk} := \mathbb{E}_i\left[Y_{i,g+k} - Y_{i,g-1} \mid \tau_i{=}g\right] - \mathbb{E}_i\left[Y_{i,g+k} - Y_{i,g-1} \mid \tau_i{>}g + k\right] = \ldots = \mathbb{E}_i\left[Y^1_{i,g+k} - Y^0_{i,g+k} \mid \tau_i{=}g\right].$$

These effects may then be aggregated across the cohorts observed at event time k:

$$\beta^k_{\text{ES}} := \underbrace{\sum_g w_{gk}\,\text{ATE}_{gk}}_{\textbf{ATET at event time } \boldsymbol{k}}, \qquad \sum_g w_{gk} = 1, \quad k \geq 0.$$

Estimator The natural estimator of each population ATE_{gk} is its sample analog; the estimates are then aggregated across cohorts for each event time k. In practice, event-study estimates are commonly obtained using a *dynamic two-way fixed-effects* specification:

$$Y_{it} = \alpha_i + \delta_t + \sum_{\substack{k=-K \\ k\neq -1}}^{L} \beta_k \mathbb{1}\{t - \tau_i = k\} + e_{it},$$

where $k = -1$ is the omitted reference period. The coefficients for $k \geq 0$ describe treatment-effect dynamics; those for $k < 0$ are pre-treatment placebo coefficients (under no anticipation and parallel counterfactual trends, they should be zero).

⚠ *Issues arising with TWFE regressions under heterogeneous effects are considered below.*

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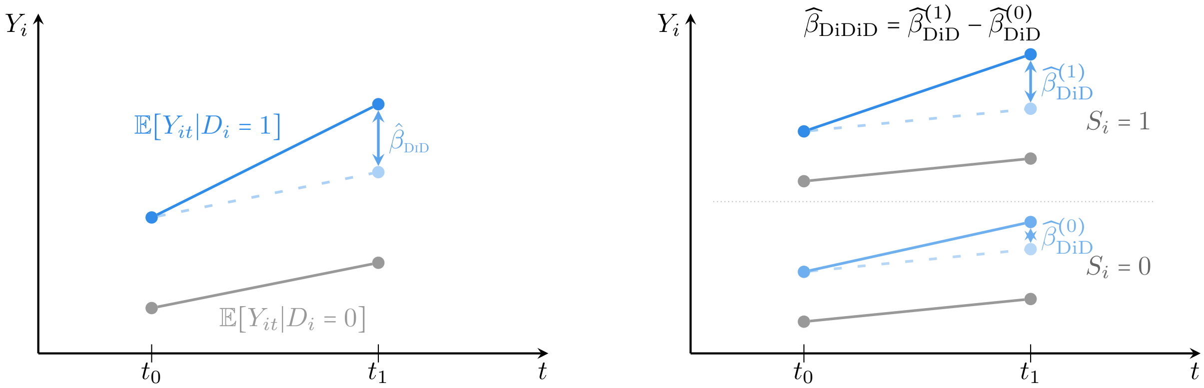
Best practices

- Plot the sequence of lead and lag estimates, with confidence intervals, of the dynamic DiD regression. For each k, the event-study estimate is the *treated-control* outcome gap in the given period relative to the *treated-control* gap in the reference period $k = -1$. Thus, for $k < 0$, these pre-treatment estimates measure whether treated and control units were already drifting apart before treatment, providing evidence about pre-trends and anticipation. Report also the joint Wald/F-test of $H_0 : \{\beta_s\}_{s<\tau-1} = 0$. However, failure to reject this null would not suffice to establish parallel counterfactual trends: the test may have low power, and perfectly parallel pre-trends do not rule out a group-specific shock occurring at the same time as treatment.
- If the composition of the (sub)groups changes over time, adjust for these changes. Interacting covariates X with P_t may help but is generally insufficient. In the case of an unbalanced panel where outcomes aren't measured periodically, we can also test for monitoring endogeneity by regressing the probablity of observing the outcome on treatment and controls.
- As in any observational study, consider adjusting for important pretreatment predictors to improve precision and reduce model sensitivity.
- *DiDiD* — A triple difference makes for a very specific control group. One should justify a DiDiD by explaining why a double difference isn't satisfactory (i.e., why the control and treatment groups in the double difference may have different counterfactual trends), and why the additional subgroup provides a credible bias correction: that is, why violations of ordinary DiD parallel trends should affect the $S_i = 1$ and $S_i = 0$ subgroups similarly.
- *DiDiD* — Examine whether the $S_i = 1$ versus $S_i = 0$ outcome gap evolves similarly across the G_i groups before treatment.

Strengths and weaknesses

- 👍 Identification only requires repeated observations, so repeated cross-sectional data suffice, as long as the sample composition does not vary over time. Balanced panel data satisfy this condition by construction.
- 👎 Parallel counterfactual trends cannot be directly tested because untreated post-treatment outcomes for the treated group are unobserved. Parallel pre-trends provide only indirect supporting evidence.
- 👎 It is difficult to rule out other things changing at the same time, i.e., unobserved confounders (time-varying unobservables correlated with both D_{it} and Y_{it}). Specifcally, event-study designs remain vulnerable to cohort-specific time-varying shocks that coincide with treatment.
- 👍 *DiDiD* — A triple difference can difference out more confounding elements (by removing bias from differential trends between the G_i groups when those trends affect the two S_i subgroups similarly).
- 👎 *DiDiD* — It requires more data and variation.



(a) DiD

(b) DiDiD

Figure 2: DiD and DiDiD treatment effects

⚠ Two-Way Fixed-Effects estimators can be unreliable with heterogeneous TEs

Assuming identification assumptions hold, the estimation of a weighted ATET is relatively simple in the ideal setting of a binary D that only switches on and is assigned at the same time for all units. However, in more general settings (where D is nonbinary, staggered, switches off...), naive extensions of the canonical method estimate quantities that can be very far from an informative ATET.

Assume that the relevant identification assumptions—including parallel trends and, where required, no anticipation—hold. In the canonical two-group, two-period setting, the DiD estimator is numerically equal to the coefficient from a TWFE regression and identifies the ATET. However, in designs with more variety in exposure to treatment (with multiple groups and periods, variation in treatment timing, treatment switching off, non-binary treatments...), the TWFE estimator may be biased **if TEs are not constant across groups or over time** (e.g., a policy becoming more or less effective). I.e., even with all confounders accounted for, $\hat{\beta}_{\text{FE}}$ is not robust to heterogeneity of TEs across groups or periods.

1. **Source of the problem**

   Consider a binary treatment that is constant within each group-period cell (g,t), and the TWFE specification $Y_{it} = \alpha_{g[i]} + \gamma_t + \beta_{\text{FE}} D_{g[i]t} + e_{it}$. Both the ATET and the TWFE estimator are weighted sums of the average treatment effects in the treated cells, with weights summing up to one. However, while the ATET weights each treated cell by its share among treated observations, $\hat{\beta}_{\text{FE}}$ weights each treated cell by a weight w_{gt} that is proportional to and of the same sign as $(D_{gt} - D_{g.} - D_{.t} + D_{..})$,[34] such that in general, $\mathbb{E}[\hat{\beta}_{\text{FE}}] \neq \beta_{\text{ATET}}$ (de Chaisemartin and D'Haultfœuille, 2020):

   $$\mathbb{E}[\hat{\beta}_{\text{FE}}] = \mathbb{E}\left[\sum_{(gt):D_{gt}=1} w_{gt}\, \text{ATE}_{gt}\right], \qquad \beta_{\text{ATET}} = \mathbb{E}\left[\sum_{(gt):D_{gt}=1} \frac{N_{gt}}{N_1}\, \text{ATE}_{gt}\right]$$

   ✓ In the canonical two-group, two-period design, there is only one treated cell, so its weight is one. More generally, with common treatment timing and a balanced panel, the TWFE weights coincide with the treated-cell-size weights, so TWFE identifies the ATET.

   ✗ In settings with varying treatment, the $\{w_{gt}\}$ may differ from $\{N_{gt}/N_1\}$. Then, heterogeneity in the cell-level average treatment effects leads to a biased $\hat{\beta}_{\text{FE}}$. Some weights may even be negative, such that $\hat{\beta}_{\text{FE}}$ may not even identify a convex combination of the underlying treatment effects.[35]

   - When D is binary and staggered in a static TWFE regression, $\hat{\beta}_{\text{FE}}$ is a weighted average of the available two-group, two-period DiD comparisons, with nonnegative weights determined by the given subsample size and treatment variance (Goodman-Bacon, 2021).[36] Some of these 2x2 DiDs misuse an early-treated group as control for a late-treated group, which may generate negative weights when treatment effects evolve after adoption.
   - When D is binary and staggered in a dynamic TWFE regression, the coefficient on a given lead or lag can be contaminated by effects from other periods, and thus need not identify the desired dynamic treatment effect under treatment-effect heterogeneity. Apparent pre-trends may arise solely from post-treatment-effect heterogeneity (Sun and Abraham, 2021).
   - When D isn't binary, $\hat{\beta}_{\text{FE}}$ may include comparisons between the outcome evolution of a group whose treatment increases more and that of a group whose treatment increases less.

[34] Where N_{gt} is the number of observations in cell (g,t), $N_1 := \sum_{(g,t):D_{gt}=1} N_{gt}$ is the total number of treated observations, and a dot subscript means the variable's average is taken over the given dimension. While the original demonstration considers a binary treatment, the results *"apply to any ordered treatment"* (de Chaisemartin and D'Haultfœuille, 2023), s.t. $\text{ATE}_{gt} = \frac{Y_{gt}^{D_{gt}} - Y_{gt}^{0}}{D_{gt}}$. Precisely, $w_{gt} = \frac{N_{gt}\tilde{e}_{gt}}{\sum_{(g't'):D_{g't'}=1} N_{g't'}\tilde{e}_{g't'}}$, where $\tilde{e}_{gt}$ is the residual from the regression of D_{gt} on group and period fixed effects. In a balanced panel, $\varepsilon_{gt} = D_{gt} - D_{g.} - D_{.t} + D_{..}$.

[35] This can even lead to a negative coefficient $\hat{\beta}_{\text{FE}}$ while the true ATEs are positive for everyone. Ex: $1.5 \times 1 - 0.5 \times 4 = -0.5$. Note that one-way FE regressions may similarly be biased for the ATET, but unlike TWFE regressions, they always estimate a convex combination of TEs.

[36] OLS gives more weight to subgroups where the FE-adjusted treatment dummy varies more. As a result, the timing of a unit's treatment determines its weight in the regression: if the unit is treated very early or very late, then it has very little variation in treatment (it is 0 almost the whole time, or 1 almost the whole time) and so receives little weight. Recall that weighting by treatment variance is always how OLS handles heterogeneity—see Section 2.1.2.

Wooldridge (2021) highlights that the cause of the problem is not the TWFE estimator per se, but its misuse: it is applied to a restrictive model, namely, that does not allow for heterogeneity in the TE.

2. **Alternatives**

   de Chaisemartin and D'Haultfœuille (2023) reviews the rapidly growing literature on this problem and highlights:

   - Diagnosis tools
     - The `twowayfeweights` command (available in R and Stata) computes the weights $\{w_{gt}\}$.
     - The `bacondecomp` command (available in R and Stata) computes the 2×2 DiD estimators and their weights in the Goodman–Bacon decomposition of $\hat{\beta}_{\text{FE}}$, for a binary staggered treatment.
   - Alternative estimators
     - Wooldridge (2021) proposes an *extended TWFE* approach, motivated by the equivalence between TWFE and a random-effects Mundlak estimator, which includes cohort-by-period interactions to allow treatment effects to vary across treatment cohorts or periods.
     - Callaway and Sant'Anna (2021) estimates group-time average treatment effects in the treated cells using only never-treated or not-yet-treated units as the comparison group, i.e., excluding already-treated units. The approach applies to staggered-adoption designs in which treatment, once adopted, remains in place.

More generally, since the source of the problem is the heterogeneity in TE, we should be thinking about how to accommodate the relevant heterogeneity in our model.

Synthetic control method

Design An intervention affects a single aggregate unit $j = 1$ from period $t = \tau$ onward, while the other units $j = 2, ..., J+1$ remain untreated. Because no single untreated unit may provide a suitable comparison, SCM constructs a *synthetic control unit*: a weighted average of the J untreated units in the donor pool. For each unit$j$, we observe a set of k pre-intervention predictors of the outcome, $X_{1j}, ..., X_{kj}$, including pre-intervention values of Y_{jt}. Let $\mathrm{w} = (w_2, \ldots, w_{J+1})'$ denote the donor weights. The synthetic control is constructed to reproduce the treated unit's pre-intervention predictors and outcome trajectory and thereby approximate the outcome it would have experienced without treatment.

Identifying assumptions

(A1) *No anticipation:* before $t = \tau$, unit 1 is unaffected by the intervention.

(A2) *Stability of the counterfactual relationship:* the synthetic control unit—which reproduces the treated unit before the intervention—would also reproduce its untreated potential outcome after the intervention: $Y_{1t}^0 \simeq \sum_{j=2}^{J+1} w_j^* Y_{jt}^0$, for $t \geq \tau$.

Estimand For each post-intervention period $t \geq \tau$, the estimand is simply the **TET at *t*:** $\beta_{\text{SCM},t} := Y_{1t}^1 - Y_{1t}^0$.

Estimator For each post-intervention period $t \geq \tau$, the SCM estimator is

$$\hat{\beta}_{\text{SCM},t} := Y_{1t} - \hat{Y}_{\text{SU},t} = Y_{1t} - \sum_{j=2}^{J+1} \hat{w}_j Y_{jt}.$$

Best practices / Steps

1. Design phase: constructing the synthetic control
   - Common approach: We choose the weights that minimize a distance between the vector of pre-treatment characteristics for the treated unit, X_1, and the matrix of pre-treatment characteristics for the donor units, X_0:

$$\begin{aligned} \widehat{\mathrm{w}}(\mathsf{V}) := \underset{\mathrm{w}}{\operatorname{argmin}} \quad & \sqrt{(\mathrm{X}_1 - \mathrm{X}_0\mathrm{w})'\mathsf{V}(\mathrm{X}_1 - \mathrm{X}_0\mathrm{w})} \\ \text{s.t.} \quad & w_j \geq 0, \quad j = 2, ..., J+1, \quad \sum_{j=2}^{J+1} w_j = 1, \end{aligned}$$

     where V is a $k \times k$ positive semidefinite weighting matrix representing the relative importance of the predictors.

     The predictors should be selected using substantive knowledge and their ability to predict Y^0, without using post-intervention outcomes. The matrix V is commonly chosen to minimize pre-intervention mean squared prediction error $\text{MSPE}_{\text{pre}} := \frac{1}{T_0} \sum_{t<\tau} \left(Y_{1t} - \sum_{j=2}^{J+1} \hat{w}_j Y_{jt}\right)^2$. When the pre-intervention period is sufficiently long, it may be divided into separate training and validation periods to reduce overfitting.
   - Alternative methods include best subset regression or LASSO and elastic nets, which may perform better in settings with a large donor pool; see Doudchenko and Imbens (2016).
2. Estimation phase
   - Report the estimated weights $\hat{w}_2, \ldots, \hat{w}_{J+1}$.
   - Report balance in the predictors (check that the synthetic and the treated units have similar values across all matching variables) and graph the treated and synthetic-control outcome trajectories and their gap.
   - Assess robustness to (i) the predictor set $X_1, ...X_k$; (ii) the choice of units $j = 2, ..., J+1$ in the donor pool. In particular, conduct leave-one-out analyses excluding, one at a time, the donor units receiving positive weight.

3. Inference phase

⚠The canonical SCM procedure is not regression-based, and so does not get generate a conventional regression-based standard error. Inference therefore requires a separate procedure.

- *A p-value* may be computed through placebo-based inference, assessing whether the treated unit's post-intervention gap is unusually large relative to placebo gaps obtained by applying SCM to untreated units:

  (a) Iteratively reassign the intervention to each donor unit and re-estimate its synthetic control.

  (b) For each unit j, calculate a test statistic, commonly the ratio of the post-treatment root mean squared prediction error to that of the pre-treatment (Abadie et al., 2010): $R_j := \frac{\text{RMSPE}_{j,\text{post}}}{\text{RMSPE}_{j,\text{pre}}}$.

  (c) Compare R_1 with the corresponding statistics for the placebo units. Using all $J+1$ units, the rank-based p-value is

  $$p := \frac{\text{rank}}{\text{total}} = \frac{1 + \sum_{j=2}^{J+1} \mathbb{1}\{R_j \geq R_1\}}{J+1}.$$

  It is also useful to plot a histogram of the statistics, and mark the treatment unit/group in the distribution to show the exact p-value associated with the model, and to plot all the placebos and highlight the treated unit's trajectory.

- *Confidence intervals* may be constructed through test inversion; see Firpo and Possebom (2018).

Strengths and weaknesses

👍 The SCM constructs the comparison unit using a data-driven procedure that combines matching on pre-intervention characteristics with a DiD-like comparison over time.

👍 Transparency

- of the fit: by showing the actual discrepancy between the treated and the synthetic units: in X with a balance table, and in pre-intervention outcomes with a graph of the two trajectories;
- of the counterfactual: the weights make explicit what each unit is contributing to the counterfactual; whereas regression also weights the data, but one cannot see the weights.

⚠The SCM does not solve the problem of subjective researcher bias. Through the choice of the covariates themselves, one can in principle select different weights.[37]

👍 It precludes extrapolation beyond the support of the data (which can occur in extreme situations with regression) by restricting the synthetic control to the convex hull of the donor units, using non-negative weights that sum to one.

👎 It requires a credible donor pool and good pre-intervention fit for a reliable estimated counterfactual.

👎 Inference requires additional assumptions and procedures and may be underpowered when the donor pool is small.

Extensions

- Synthetic Difference-in-Differences (Arkhangelsky et al., 2021)
- Relaxing functional form assumptions. For example, unclear whether one should assume that it is the *percentage change* or the *absolute change* in the level of average outcomes over time that would have been the same across groups in the absence of treatment. For the case of repeated cross-sectional data, Athey and Imbens (2006) proposes a nonlinear version of the difference-in-differences model, called *changes-in-changes*, which does not rely on functional form assumptions, while still allowing the effects of time and treatment to vary systematically across individuals. It provides the distribution of outcomes rather than the average effect of the treatment.

[37] Ferman et al. (2020) finds possibilities for specification searching in SCM. The authors consider a variety of commonly used SC specifications, run a randomization inference test to calculate empirical p-values, and find that the probability of falsely rejecting the null in at least one specification for a 5% significance test can be as high as 14%. I.e., it is theoretically possible to 'hack' the analysis in order to find statistical significance that suits one's priors, by specification searching. It is recommended to present multiple results under a variety of commonly specified specifications.

Summary of canonical identification strategies and regression-based consistent estimators

Method	Identifying assumptions	Estimand β and corresponding TE	Consistent[38] estimator $\hat{\beta}$	Strengths and weaknesses
RCT	– independence	$\beta := \mathbb{E}[Y_i \mid D_i{=}1] - \mathbb{E}[Y_i \mid D_i{=}0] = \ldots = \underbrace{\mathbb{E}[Y_i^1 - Y_i^0]}_{\text{ATE}}$	$\hat{\beta}_{\text{OLS}}$ of the regression $Y_{it} = \alpha + \beta\ D_i + e_{it}$	👍 random assignment gives (A) ⇒ RCT = *"gold standard"*
Selection on observables	– conditional independence – overlap in X	$\beta := \sum_x w(x)\big(\mathbb{E}[Y_i \mid D_i{=}1, X_i{=}x] - \mathbb{E}[Y_i \mid D_i{=}0, X_i{=}x]\big)$ $= \ldots = \underbrace{\sum_x w(x)\mathbb{E}[Y_i^1 - Y_i^0 \mid D_i{=}1, X_i{=}x]}_{\text{ATET}},$	$\hat{\beta}_{\text{OLS}}$ of the regression $Y_{it} = \alpha + \beta\ D_i + f(X_i) + e_{it}$	👎 must observe all confounders
IV	– exclusion restriction – independence – relevance – monotonicity	$\beta_{\text{IV}} := \dfrac{\text{cov}[Y_i, Z_i]}{\text{cov}[D_i, Z_i]} = \ldots = \dfrac{\mathbb{E}[Y_i \mid Z_i{=}1] - \mathbb{E}[Y_i \mid Z_i{=}0]}{\mathbb{E}[D_i \mid Z_i{=}1] - \mathbb{E}[D_i \mid Z_i{=}0]}$ $= \ldots = \underbrace{\mathbb{E}\left[Y_i^1 - Y_i^0 \mid D_i^1{=}1, D_i^0{=}0\right]}_{\text{LATE for compliers}}$	$\hat{\beta}_{\text{2SLS}}$ of the two-stage regression $D_i = \delta + \gamma\ Z_i + v_i$ $Y_i = \widetilde{\alpha} + \widetilde{\beta}\ \widehat{D}_i + e_i$	👍 reliable exogenous variation 👎 **biased**, bias ↑ w. Z weakness 👎 precision ↓ w. Z weakness
Sharp RD	– local continuity – relevance	$\beta_{\text{RD}} := \lim\limits_{x \to c^+} \mathbb{E}[Y_i \mid X_i{=}x] - \lim\limits_{x \to c^-} \mathbb{E}[Y_i \mid X_i{=}x]$ $= \ldots = \underbrace{\mathbb{E}[Y_i^1 - Y_i^0 \mid X_i{=}c]}_{\text{LATE at the cutoff}}$	$\hat{\beta}_{\text{OLS}}$ of the regression $Y_i = \alpha + \beta D_i + f(X_i, \phi_l) + D_i f(X_i, \phi_r) + e_i,$ with choice of flexible $f()$	👍 ≈ local randomized experiment 👎 **biased**, bias ↑ w. bandwidth 👎 risk of low power 👎 risk of low external validity
Fuzzy RD	– exclusion restriction – local continuity – relevance – local monotonicity	$\beta_{\text{RD}} := \dfrac{\lim\limits_{x \to c^+} \mathbb{E}[Y_i \mid X_i{=}x] - \lim\limits_{x \to c^-} \mathbb{E}[Y_i \mid X_i{=}x]}{\lim\limits_{x \to c^+} \mathbb{E}[D_i \mid X_i{=}x] - \lim\limits_{x \to c^-} \mathbb{E}[D_i \mid X_i{=}x]}$ $= \ldots = \underbrace{\mathbb{E}[Y_i^1 - Y_i^0 \mid X_i{=}c]}_{\text{LATE for compliers at the cutoff}}$	$\hat{\beta}_{\text{2SLS}}$ of the two-stage regression $D_i = \delta + \gamma Z_i + f(X_i, \phi_l) + Z_i f(X_i, \phi_r) + v_i$ $Y_i = \alpha + \boldsymbol{\beta} \widehat{D}_i + f(X_i, \widetilde{\phi}_l) + Z_i f(X_i, \widetilde{\phi}_r) + e_i$ with choice of flexible $f()$	*Combination of the strengths and weaknesses of IV and Sharp RD*
DiD	– ∥ counterfactual trends across groups – no anticipation – stable group compositions	$\beta_{\text{DiD}} := (\mu_{G_1 P_1} - \mu_{G_1 P_0}) - (\mu_{G_0 P_1} - \mu_{G_0 P_0})$ $= \ldots = \underbrace{\mathbb{E}[Y_{i1}^1 - Y_{i1}^0 \mid G_i{=}1]}_{\text{ATET in post period}}$	$\hat{\beta}_{\text{OLS}}$ of the regression $Y_{it} = \beta\ G_i P_t + \lambda_G + \lambda_P + e_{it}$	👍 differences out unobserved time-invariant confounders
DiDiD	– ∥ counterfactual trends in subgroup differences – no anticipation – stable group × subgroup compositions	$\beta_{\text{DiDiD}} := \left[(\mu_{G_1 S_1 P_1} - \mu_{G_1 S_1 P_0}) - (\mu_{G_1 S_0 P_1} - \mu_{G_1 S_0 P_0})\right]$ $- \left[(\mu_{G_0 S_1 P_1} - \mu_{G_0 S_1 P_0}) - (\mu_{G_0 S_0 P_1} - \mu_{G_0 S_0 P_0})\right]$ $= \ldots = \underbrace{\mathbb{E}\left[Y_{i1}^1 - Y_{i1}^0 \mid G_i{=}1, S_i{=}1\right]}_{\text{ATET in post period}}$	$\hat{\beta}_{\text{OLS}}$ of the regression $Y_{it} = \beta\ G_i S_i P_t + \lambda_{GS} + \lambda_{GP} + \lambda_{PS} + e_{it}$	👍 differences out more potential confounders than in DiD 👎 requires more variation
Event-study	– ∥ counterfactual trends across treatment cohorts – no anticipation – stable cohort compositions	$\forall k,\ \beta^k_{\text{ES}} := \sum_g w_{gk} \big(\mathbb{E}[Y_{i,g+k} - Y_{i,g-1} \mid \tau_i{=}g]$ $- \mathbb{E}[Y_{i,g+k} - Y_{i,g-1} \mid \tau_i{>}g{+}k]\big)$ $= \ldots = \underbrace{\sum_g w_{gk}\, \mathbb{E}[Y^1_{i,g+k} - Y^0_{i,g+k} \mid \tau_i{=}g]}_{\text{ATET at event time } k}$	$\{\hat{\beta}^k_{\text{OLS}}\}_k$ of the regression $Y_{it} = \alpha_i + \delta_t + \sum_{\substack{k=-K \\ k \neq -1}}^{L} \beta_k\, \mathbb{1}\{t - \tau_i = k\} + e_{it}$	👍 flexible 👎 difficult to rule out unobserved confounders
SCM	– no anticipation – counterfactual stability	$\beta_{\text{SCM},t} := \underbrace{Y^1_{1t} - Y^0_{1t}}_{\text{TET at time } t \geq \tau}$	$\hat{\beta}_{\text{SCM},t} := Y_{1t} - \hat{Y}_{\text{SU},t} = Y_{1t} - \sum_{j=2}^{J+1} \hat{w}_j Y_{jt}$	👍 transparent procedure and fit 👎 inference requires addit. steps

[38]When the estimand is an ATET, $\hat{\beta}_{\text{OLS}}$ is generally inconsistent under heterogeneous effects as its weights aren't the ATET weights: $w_{\text{OLS}}(x) := \mathbb{V}[D_i \mid X_i{=}x]\,\Pr(X_i{=}x) \neq \frac{\Pr(D_i=1 \mid X_i=x)}{\Pr(D_i=1)}\Pr(X_i{=}x)$.

4 Analysis stage: steps for stronger causal inferences

4.1 Identification strategies only provide so much

Strong identification strategies provide credible counterfactual comparisons that identify a causal estimand, for which we can define and estimate an unbiased estimator, often through regression. This remains limited, in at least three important ways:

1. Under the CIA and overlap, causal effects can be consistently estimated either by balancing the covariate distributions of treated and untreated units, through matching or weighting, or by correctly modeling the conditional response surfaces $\mathbb{E}[Y_i \mid D_i{=}1, \mathrm{X}_i]$ and $\mathbb{E}[Y_i \mid D_i{=}0, \mathrm{X}_i]$. One strategy is therefore to focus on appropriately adjusting for treatment assignment, thereby limiting reliance on unknown response-surface functional forms; another is to model those response surfaces directly.[39] Doubly robust estimators combine an outcome model and a propensity-score model and remain consistent if either is correctly specified; see Imbens and Wooldridge (2009, sec. 5.6) and Hill (2011, section 2).

2. An estimator $\hat{\theta}$ being unbiased means that its sampling distribution—over repeated samples or trials of the same size—is centered around the true value of the estimand θ. Its realization in a particular study may nevertheless be far from θ, especially when the sample is small. Adjusting for prognostic pre-treatment covariates can account for realized chance imbalance between groups and improve precision.

3. What knowledge is generated by an estimate of the ATET? Identification strategies are generally—and this document is no exception—motivated by the RCT as the gold standard. When treatment represents a well-defined intervention, its average effect on the treated may itself be the principal quantity of interest. In other contexts, however, estimating the magnitude of an effect without identifying the mechanisms producing it may provide limited insight and does not establish whether the effect generalizes to other populations or settings.[40]

The rest of this section describes analysis-stage steps that may address these limitations and generate more informative inferences. Specifically:

1. *pre-estimation:* restructuring the data to improve overlap and covariate balance;
2. *estimation:* including appropriate covariates and accommodating treatment-effect heterogeneity;
3. *post-estimation:* assessing modeling assumptions and external validity.

4.2 *Pre-estimation:* Restructuring

Causal identification relies on comparisons intended to recover unobserved counterfactual outcomes. While the relevant comparison differs across designs—for example, between treated and untreated units, instrument groups, observations around a cutoff, or treatment cohorts over time—overlap and balance in relevant pre-treatment covariates can make all these comparisons more precise and less dependent on modeling assumptions.

Two problems may arise:

- **Incomplete overlap:** the *support* of the distribution of X differs across the groups. Some treated observations have no comparable control observations.
  ↪ *In regions without overlap, estimating counterfactual outcomes requires extrapolation, so inferences depend on modeling assumptions rather than direct comparisons in the data.*

- **Imbalance:** the *shape* of the distribution of X differs across the groups; e.g., the mean or skewness...
  ↪ *This may increase model dependence, imprecision, and make the identifying assumptions less credible.*

[39] Recall Angrist and Pischke (2008)'s argument that linear regression is always relevant in that the linear predictor $\mathrm{X}\hat{\beta}$ provides the best linear approximation to the conditional expectation function, $\mathbb{E}[Y_i \mid D_i, \mathrm{X}_i]$. However, if the response surfaces are strongly nonlinear, a global linear approximation may be very poor and provide a very biased estimate of the causal effect, even when local linearity is reasonable (Imbens and Wooldridge, 2009). Note that functional-form restrictions—and hence potential misspecification—could in principle be avoided by saturating the model, i.e., estimating a separate conditional mean for every combination of treatment and covariate values. This is rarely tractable with many or continuous covariates.

[40] ⚠ Mechanisms cannot generally be identified by simply conditioning on intermediate outcomes: doing so changes the estimand and may introduce post-treatment bias. Identifying direct and indirect effects requires additional assumptions; see the following subsection.

Restructuring to balance the observed covariates The less the treatment and control groups have overlap and balance with respect to confounders X, the more inferences rely on the model instead of on data, and so aren't robust to model misspecification. On the contrary, if the distributions of X are similar across the groups, then, even if we misspecify the form of the relationship, we should still get a reasonable estimate of the treatment effect (Gelman et al., 2020). This motivates *restructuring* the sample prior to estimation by selecting, matching, or weighting observations to improve overlap and balance.

⚠ *Matching does not itself provide identification; it merely strengthens the empirical comparison on which the identification strategy relies. A causal interpretation requires conditional ignorability,* $(Y_i^1, Y_i^0) \perp\!\!\!\perp D_i \mid \mathrm{X}_i$*, and overlap. These assumptions being satisfied through matching would require the observed* X *to capture all relevant confounding and cannot be tested from the observed data. Therefore matching is not an alternative to a design-based method. It is an adjustment strategy, not an identification strategy.*

The appropriate restructuring depends on the estimand and design. Consider the example target of the ATET. We can restructure the control group so that its distribution of observed relevant covariates resembles that of the treated group. In that context, matching can be used in two ways:

- **In place of regression: matching as an estimation method**

  The 'regression with controls' estimand and the nonparametric 'covariate-matching' estimand are two different ways of balancing the X (Angrist and Pischke, 2008).[41] In practice, a matching estimator is obtained by comparing each treated unit with control unit(s) having similar observed X, and averaging the resulting outcome differences using weights corresponding to the target estimand. Computing standard errors for inference must account for any repeated use of control observations.

- **On top of regression: matching as a preprocessing method**

  Matching is used to construct more comparable groups and followed by regression for further adjustment, modeling treatment-effect heterogeneity, or gains in precision. Used as nonparametric preprocessing, matching reduces the dependence of the subsequent regression on functional-form assumptions (Gelman et al., 2020; Ho et al., 2007).

Common distance metrics Matching is relatively straightforward with a single continuous observed confounder X_1, or even with an additional binary X_2 (e.g., matching on X_1 within strata defined by X_2). It becomes more difficult as the number of covariates increases. One approach is to summarize multidimensional differences in X using a univariate distance metric, and match each treated unit to the nearest control unit according to that metric:

- **Mahalanobis distance**

  The Mahalanobis distance between observations i and j is defined as

  $$d_{ij} := \sqrt{(\mathrm{X}_i - \mathrm{X}_j)' \Sigma_{\mathsf{X}}^{-1} (\mathrm{X}_i - \mathrm{X}_j)},$$

  where Σ_{X} is the sample covariance matrix of the covariates. The metric is invariant to the scale of the covariates and accounts for their correlation structure.

- **Propensity score**

  The propensity score approach uses as distance metric between units the difference in their estimated propensity score or conditional probability of treatment:

  $$d_{ij} := |\hat{p}(\mathrm{X}_i) - \hat{p}(\mathrm{X}_j)|, \quad \text{where } p(\mathrm{X}_i) := \Pr(D_i = 1 | \mathrm{X}_i)$$

  If treatment assignment is conditionally ignorable given X, it is also conditionally ignorable given the propensity score: $(Y_i^1, Y_i^0) \perp\!\!\!\perp D_i \mid \mathrm{X}_i \implies (Y_i^1, Y_i^0) \perp\!\!\!\perp D_i \mid p(\mathrm{X}_i)$, provided sufficient overlap. Conditioning appropriately on the *true* propensity score therefore identifies the same treatment effects as conditioning on the full set of observed X, hence the appeal of the propensity score apporach. However,

[41] While neither the regression nor the matching *estimands* give any weight to covariate cells that don't have both treated and control observations, the regression *estimator* uses modeling assumptions that implicitly involve extrapolation across cells, such that cells without overlap can end up contributing to the estimate. Exact matching instead avoids such extrapolation; while nearest-neighbor matching may produce distant matches when overlap is weak, matching in general can make the absence of overlap explicit and limit extrapolation.

this is a population property; matching approximately on an estimated score need not produce exact covariate balance in a finite sample.

See Rosenbaum and Rubin (1983) for the propensity-score approach and the balancing-score theorem.

Algorithm for propensity-score matching:

1. *Estimate the propensity score:* model $p(\mathrm{X}_i) = \Pr(D_i = 1 \mid \mathrm{X}_i)$, e.g., with a logistic regression model of D_i on X_i, using only pre-treatment covariates, and obtain $\hat{p}_i$.
2. *Match units:* restrict the analysis to the region of common support and choose a matching procedure, such as nearest-neighbor(s) matching, coarse matching by quintile blocks of $\hat{p}_i$, with or without replacement.[a]
3. *Assess overlap and balance:*
   - overlap: compare the distributions of $\hat{p}_i$ across the treatment and control groups;
   - balance: compare the distribution of each X across the groups before vs after matching.

   If balance is inadequate, revise the propensity-score specification or the matching procedure without examining the outcome (e.g., try a less coarse matching procedure, or a richer model with interactions or higher order terms of covariates).
4. *Estimate the ATET using the restructured data:* One can compute either a matching estimator or a regression estimator:
   - matching: compute the appropriately weighted difference in mean outcomes between treated units and their matched controls; Example: Almond et al. (2005) presents both an OLS and a matching estimator.
   - regression: fit a regression of Y on:
     * D, $\hat{p}$ and $D \times \hat{p}$. See Doug Almond's "Algorithm for Estimating the Propensity Score" pdf (from Ken Chay's 2001 UC Berkeley econometrics class); or
     * D and X. See Gelman et al. (2020, ch. 20): *"This gives us an additional chance to adjust for differences in distributions of X that typically remain between the groups (to decrease bias and increase efficiency). The overlap and balance created by the matching should make this model more robust to potential model misspecification; that is, even if this model isn't quite right (for example, excluding a key interaction, or assuming linearity when the underlying relation is strongly nonlinear) our coefficient estimate should still be close to correct, conditional on ignorability being satisfied."*[b]

[a] Matching with vs without replacement is akin to a bias-variance trade-off: matching with replacement should yield better matches on average, therefore better balanced and less biased estimators; however, it can result in over-using certain units or ignoring other close matches, i.e., missing out on important information in the data, and potentially increasing the variance of the estimators.

[b] ⚠The standard errors from this regression are not technically correct, as: (1) matching induces correlation among the matched observations—the regression model, however, if correctly specified, should account for this by including the variables used to match; (2) the fact that the propensity score has been estimated from the data is not reflected in the calculations (Gelman et al., 2020, p. 404).

4.3 *Estimation:* Regression controls and TE heterogeneity

Required/bias-inducing regression controls (for bias) The identification strategy determines which variables must and must not be adjusted for:

✓ Adjust for the confounding that threatens the identification strategy,[42] by controlling for a set of pre-treatment non-colliders that blocks all open back-door paths between D and Y.[43]

⚠More controls is not always better! Controlling for some confounding paths but not *all* can sometimes make the situation worse by amplifying the bias from an omitted confounder. Middleton et al. (2016) demonstrates such bias amplification in the simple case of two confounders. Suppose that, for centered

[42] Blocking all $D - Y$ back-door paths by regression adjustment is not required under every identification strategy. IV, RD, DiD, or randomization can identify an effect despite unblocked confounding between D and Y.

[43] A back-door path is any path between D and Y that begins with an arrow into D. This includes reverse-causality paths such as $D \leftarrow \cdots \leftarrow Y$. Here, we assume that no such paths exist, so all back-door paths are confounding paths.

variables, $Y_i = \alpha + \beta D_i + \delta A_i + \gamma B_i + e_i$, where A and B are confounders, $A \perp\!\!\!\perp B$, but only A is observed. Should we include A?

- Omitting both A and B yields the asymptotic bias $\left(\text{plim}\,\hat{\beta}_{Y|D} - \beta\right) = \underbrace{\delta\frac{\text{cov}[D,A]}{\mathbb{V}[D]}}_{:=b_A} + \underbrace{\gamma\frac{\text{cov}[D,B]}{\mathbb{V}[D]}}_{:=b_B}$.
- Omitting B but including A yields the asymptotic bias $\left(\text{plim}\,\hat{\beta}_{Y|D,A} - \beta\right) = \frac{1}{1-R^2_{D|A}}\,b_B$, where $R^2_{D|A}$ is the R-squared of the linear regression of D on A.

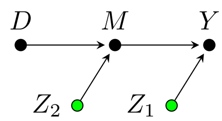
⇒ When B is omitted, including A increases net bias iff $\frac{|b_B|}{1-R^2_{D|A}} > |b_A + b_B|$.

Instruments are the canonical example of pure bias amplifiers. For a valid instrument A, $\delta = 0$ and hence $b_A = 0$: conditioning on it magnifies the bias from B by the factor $1/(1 - R^2_{D|A})$. Middleton et al. (2016) shows that also group fixed effects can be pure bias amplifiers, e.g., when the group-level structure in Y does not covary with the group-level structure in D—and so aren't necessarily a harmless way of accounting for group-level confounding.[44]

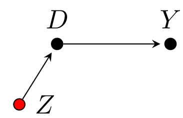
✗ Do *not* adjust for variables affected by the treatment when they are also determining Y—mediators—or determined by Y—colliders.[45] Conditioning on a mediator blocks part of the causal effect, and conditioning on a collider may open a noncausal path between D and Y; either biases the estimate of the total causal effect.

Optional good/bad regression controls (for efficiency) Separately from those dictated by the identification strategy, which additional pre-treatment covariates should we adjust for?
As a general rule of thumb, among variables that do not interfere with identification, adjusting for predictors of Y improves precision, whereas adjusting for variables that predict D but not Y reduces it. Cinelli et al. (2021) provides a detailed and nuanced description of the different cases using DAGs.

👍 Adjusting for *pre*-treatment covariates strongly associated with Y—whether a direct association like Z_1, or a mediated association like Z_2—can reduce the residual variance of Y and therefore the standard error of $\hat{\beta}$, even when these covariates are unrelated to D in the population. It may also correct realized chance imbalance. This applies also to completely randomized experiments.



👎 Adjusting for determinants of D will instead reduce the variation of D. This may reduce the precision of the linear estimator $\hat{\beta}$ in finite samples, i.e., increase its asymptotic variance; and, if residual confounding remains, amplify its bias.



Given this mess, should we avoid controls altogether and regress only on D?
→ Not generally. The unadjusted estimate provides a useful design-based benchmark, but appropriately chosen pre-treatment controls can improve precision and account for observed random or systematic imbalance. The relevant comparison is not controls versus no controls, but substantively justified controls versus indiscriminate adjustment.

- Rubin (1974) argues for considering controls: *"The investigator should be prepared to consider the possible effect of other variables besides those explicit in the experiment. Often additional variables will be ones that the investigator considers relevant because they may causally affect Y; therefore, [they] may want to adjust the estimate $\Delta\bar{Y}$ and significance levels of hypotheses to reflect the values of these variables in the study. [...] An investigator who refuses to consider any additional variables is in fact saying that [they] do not care if $\Delta\bar{Y}$ is a bad estimate of the typical causal effect of the treatment but instead is satisfied with mathematical properties* [i.e., unbiasedness] *of the process by which [they] calculated it."*
- Gelman et al. (2020, p.368) further nuances the dichotomy between benefits in terms of bias versus precision: *"Under a clean randomization, adjusting for pre-treatment predictors in this way does not change what we*

[44] Their harmlessness refers to bias, while they may be inefficient. Fixed effects lead to comparing much smaller changes than if we were to look at the entire range of data. If variables are measured rather imprecisely, we will be removing a lot of the signal but not any of the noise, therefore the power of the analysis goes down. See `http://www.g-feed.com/2012/12/the-good-and-bad-of-fixed-effects.html`

[45] There are *sometimes* things we can learn from a regression with bad controls; see `http://www.g-feed.com/2012/10/bad-control.html` (last ¶), referring to Maccini and Yang (2009).

are estimating. However, if the predictor has a strong association with the outcome it can help to bring each estimate closer (on average) to the truth, and if the randomization was less than pristine, the addition of predictors to the equation may help us adjust for systematically unbalanced characteristics across groups. Thus, this strategy has the potential to adjust for both random and systematic differences between the treatment and control groups (that is, to reduce both variance and bias), as long as these differences are characterized by differences in the pre-test."

Treatment effect heterogeneity Treatment effects may be heterogeneous.[46] We could therefore relax the restrictive assumption of a constant effect and model how it varies. There are multiple ways of doing so:

- If we expect the treatment effect to vary with a pre-treatment covariate X, we may interact D with X:

  $$Y_i = \alpha + \beta D_i + \gamma X_i + \delta D_i X_i + u_i.$$

  The conditional treatment effect is then $\beta + \delta X_i$.[47] Gelman et al. (2020) notably recommends interacting treatment with covariates that are strong predictors of Y.

- If the data are hierarchical, with units i nested in groups $g[i]$, we may let the treatment effect vary by group and pool the group-specific effects $\{\beta_g\}$:
  - *Full pooling.* Impose $\beta_g = \beta$, ignoring effect heterogeneity between groups.
  - *No pooling.* Estimate independent β_g, for example by interacting D with group indicators, with the risk of overstating the variation between groups, i.e., overfitting the data.
  - *Partial pooling.* The slope coefficients vary by group, and are themselves given a probability model: $\beta_g \sim \mathcal{F}(\mu_\beta, \sigma_\beta)$. I.e., we *model* the variation between groups inside a multilevel model. The group-level model may itself have predictors or not. The group effects are estimated jointly and pulled toward their predicted group-level mean $\hat{\mu}_\beta$, more strongly for smaller or noisier groups. Partial pooling is thereby a compromise between the extremes of full pooling ($\sigma_\beta \to 0$) and no pooling ($\sigma_\beta \to \infty$).

⚠ **Don't overfit: regularize** Adding predictors and interactions increases sampling variability and the risk of fitting noise. Complexity can be constrained through, for example:

- partial pooling (multilevel modeling);
- informative or weakly informative priors (Bayesian inference);
- LASSO;
- elastic net regression, which combines LASSO and ridge penalties, such that we minimize the sum of squared residuals plus a penalty term:

  $$\hat{\beta} = \arg\min_{\beta} \left\{ \text{SSR} + \lambda \left[\alpha \sum_p |\beta_p| + \tfrac{1-\alpha}{2} \sum_p \beta_p^2 \right] \right\}.$$

  When $\lambda = 0$, this reduces to OLS; $\alpha = 1$ gives LASSO and $\alpha = 0$ gives ridge regression. Elastic net is particularly useful when predictors are numerous and correlated. The tuning parameters (α, λ) should be selected using an explicit criterion, such as cross-validation, or one may consider reporting the robustness of the estimated treatment effect to different values of these parameters.

4.4 *Post-estimation:* Supporting assumptions and external validity

4.4.1 Diagnosis tests of modeling assumptions

See `https://clairepalandri.github.io/docs/microeconometrics.pdf`.

[46] For simplicity, let's assume that we are interested in *average* treatment effects, and so focus on variation in the first moment (mean) of the outcome distribution. We could also consider TE variation as variation in the second moment (variance) or in the overall outcome distribution (quantiles). Feller and Gelman (2015) provides some discussion of these cases.

[47] If X is continuous, center it, such that β is the effect at the mean of X. Note that a problem when X is continuous X is that we rarely have a reason a priori to make a particular assumption of the parametric form of the interaction. Some ways to get around this include using flexible models such as bins (although this generates another specification search: choosing cutpoints) or splines.

4.4.2 Falsification tests of identifying assumptions

Identifying assumptions cannot generally be tested directly. However, we can run balance checks and placebo analyses to assess some of their observable implications. Failures then weaken confidence in identification; passing increases confidence but does not establish its validity.

Show balance in pre-treatment covariates across groups Causal inference requires the treated and untreated groups to be comparable in the dimensions relevant to the identification strategy—possibly conditionally on some covariates.
In an RCT, random assignment implies that the distributions of pre-treatment covariates are equal across groups *in expectation*; realized differences may occur by chance. RCT papers therefore typically report a *balance table* describing these differences.[48] In observational settings, one should similarly document the balance of $\bar{X}$ across treatment status.

- Imbens and Wooldridge (2009, eq. 3) suggests reporting the normalized difference in means, which is a scale-free distance metric, where S_0^2 and S_1^2 are the sample variances in the control and treatment groups:[49]

$$\widetilde{\Delta}X := \frac{\bar{X}_1 - \bar{X}_0}{\sqrt{(S_0^2 + S_1^2)/2}}.$$

- Gelman et al. (2020) suggests plotting:
  - for a binary X, the raw difference in means, $\Delta X := \bar{X}_1 - \bar{X}_0$. *"Since the standard deviation for binary variables is determined by exclusively the mean it could be misleading to standardize."*
  - for a continuous X, the standardized difference in means, where S_{ref} is the standard deviation in the inferential group:

$$\widetilde{\Delta}X := \frac{\bar{X}_1 - \bar{X}_0}{S_{\text{ref}}}.$$

Falsification tests The general approach to support core identifying assumptions is to show that the specification does not find an effect when one indeed *should not* exist, e.g., by applying the design to a relevant placebo outcome, exposure, population, date, or cutoff. If the analysis picks up an effect where there isn't one, it suggests confounding, another violation of the identifying assumptions, or model misspecification.[50]

- **IV**
  - *Relevance* is directly assessable from the first-stage relationship between Z and D.
  - *Exclusion* is not directly testable. One can nevertheless test whether Z predicts Y in a setting where D can't affect Y, e.g., using an alternative population or outcome that can't be affected by the treatment. Finding a reduced-form relationship between Z and Y would mean Z is related to Y through another channel than D, falsifying the exclusion restriction.
- **RD**
  - *Local continuity* cannot be tested directly, but implies several observable diagnostics:
    * Predetermined covariates and pre-treatment placebo outcomes should not jump at the cutoff c. Such jumps would suggest that other determinants of Y are also discontinuous at c.[51]

[48] One should show for each X the difference of sample means between the two groups, but not a t-test of whether that difference is significantly different from zero. Indeed, a t-test tells us whether that difference could have happened by chance. The random assignment already guarantees that any difference observed would have happened by chance (unbiasedness). A t-test in this context is therefore conceptually unsound. Hayes and Moulton (2017) explain that *"the point of displaying between-arm comparisons is not to carry out a significance test, but to describe in quantitative terms how large any differences were, so that the investigator and reader can consider how much effect this may have had on the trial findings."* Angrist and Pischke (2008, table 2.2.1, p.14) ~~seem to disregard that in discussing the STAR experiment *"The P-value in the last column is for the F-test of equality of variable means across all three groups. [...] Differences in these characteristics across the three class types are small and none are significantly different from zero. This suggests the random assignment worked as intended."*~~

[49] ⚠ This is different from the t-statistic for the null hypothesis of equal means: $t = (\bar{X}_1 - \bar{X}_0)/\sqrt{S_0^2/N_0 + S_1^2/N_1}$.

[50] Falsification tests should not be confused with robustness checks, which estimate alternative specifications testing the same hypotheses.

[51] This test of the local randomization assumption is similar to the testing of the global randomization assumption using a balance table of baseline covariates in a randomized experiment.

    * Units may precisely sort around the cutoff. In that case, we may expect some bunching of units at the cutoff, i.e., a discontinuity at c. McCrary (2008) proposes testing whether the density of the running variable is continuous at c.
    * Outcomes should not jump at placebo cutoffs $\tilde{c}$. Such jumps would suggest natural discontinuities in Y or misspecification of the functional form. Imbens and Lemieux (2008) suggests separately taking each side of the true cutoff, using its median running-variable value as a placebo cutoff, and checking that there is no discontinuity in Y.
  - *Relevance and local support* can be tested by
    (i) verifying that treatment jumps at c (in a fuzzy RD, it is the estimated first-stage discontinuity; in a sharp RD, it follows from the assignment rule, though it should still be checked);
    (ii) inspecting whether sufficiently many observations exist close to c on both sides.
  - *Exclusion restriction [fuzzy RD]*. Post-treatment placebo outcomes that cannot be affected by D should not jump at c.[52]

- **DiD**
  - *Parallel counterfactual trends* cannot be tested directly, but relevant falsification analyses include:
    * comparing pre-treatment trends;
    * using placebo outcomes that can't be affected by D but could by plausible confounders;
    * assigning treatment to earlier placebo dates.

    Ex. Linden and Rockoff (2008) estimates the effect of the arrival of a registered sex offender at time τ on nearby property prices: $D_{it} = G_i P_t$, with $G_i := \mathbb{1}\{\text{distance} \leq 0.1 \text{ mile}\}$ and $P_t := \mathbb{1}\{t \geq \tau\}$. If the prices of nearby and comparison properties followed different counterfactual trends, i.e., were trending over time differently regardless of the arrival, the estimate would be spurious. The authors assign earlier placebo arrival dates and find no effect.
  - *Stable group composition over time* is guaranteed with panel data, unless attrition, entry, or missing observations change the observed sample. In repeated cross-sections, one may compare the covariate distributions of each group over time, using regression.

- **SCM**
  - *In-time placebos* or *backdating*: assign treatment to an earlier pre-treatment date and check that no comparable gap emerges. Abadie et al. (2015), for example, backdates the intervention and re-estimates the synthetic control using only earlier data.
  - *In-space placebos*: successively reassign treatment to donor-pool units and compare their post-treatment gaps with that of the treated unit, accounting for differences in pre-treatment fit.

Robustness to the assumption of no unobserved confounders Rather than arguing whether zero confounding is plausible or not, we could assess the amount of confounding necessary to change our conclusions. (Cinelli and Hazlett, 2020) proposes to report, alongside the estimate and confidence interval, the *robustness value*: the minimum strength of association an omitted confounder would need to have with both D and Y, conditional on the included covariates, to overturn a specified conclusion. The R package `sensemakr` implements these sensitivity analyses for OLS.

[52] Note that a jump would suggest either a direct effect of crossing the cutoff or a violation of local continuity; the two explanations cannot generally be distinguished by this test alone.

5 Presentation

Validity of a statistical analysis

- **Internal validity** is the extent to which the causal effect *for the population being studied* is correctly identified and estimated.
  It depends on how well the study rules out alternative explanations for the estimated relationship. Threats include confounding, reverse causality, sample selection, measurement error, violations of the identifying assumptions, model misspecification...
- **External validity** is the extent to which the estimated causal effect can be generalized to other populations, settings, or versions of the treatment.
  ⚠ *Even in randomized trials, the experimental sample may differ from the target population. Reweighting can address differences explained by observed variables, but not necessarily selection based on unobservables.*

5.1 Characterizing and assessing the empirical strategy

For any econometric analysis aiming at causal inference, one should characterize the following elements and assess whether each is credible:

1. Research question and estimand – *What causal effect is being estimated, for which population and over what time horizon?*
2. Ideal experiment – *What hypothetical experiment would identify this effect?*
   This requires defining the corresponding counterfactual comparison: *How would the outcomes of the relevant units differ under alternative treatment states?*
3. Identification strategy – *How are the available data used to construct comparisons that approximate those generated by the ideal experiment?* Specify the identifying assumptions, the arguments supporting their plausibility, and the particular effect identified (ATE, ATET, LATE, etc.). Consider credible scenarios under which these assumptions might fail.
4. Estimation and inference methods – *How is the identified effect estimated, and how is uncertainty quantified?* Assess the functional-form, weighting, sampling, and dependence assumptions involved.
5. Diagnostic, falsification, and sensitivity analyses – *What observable implications of the identifying assumptions can be examined, and how sensitive are the results to reasonable alternative specifications?*

All these items can be planned before opening the dataset. The multiple descriptions and checks help support the internal validity of the analysis.

5.2 Putting the paper in perspective

In addition to assessing the empirical strategy, one may want to discuss:

- Contributions to the substantive literature
- Methodological contributions
- Interpretation, relevance, and scope of the results
  - w.r.t. policy – *Do the analyses performed address the policy question of interest?*
  - w.r.t. the literature – *How does the paper account for its results compared to others in the literature?*
  - w.r.t. external validity– *To what extent are the results likely to generalize to other populations, settings, or interventions?*

6 Other branches of causal modeling

6.1 *Which uncertainty matters?* Randomization inference (RI)

"In randomization-based inference, uncertainty arises naturally from the random assignment of the treatments, rather than from the hypothesized sampling from a large population." (Athey and Imbens, 2017)

Common frequentist inference techniques are often justified from a *sampling-based* perspective. The units are treated as random draws from a larger population, and one asks: *What would have occurred under a different random sample?*

In causal inference studies, there is also another type of variation at play: variation in *assignment of treatment*, i.e., *design-based* uncertainty corresponding to what the regression outcome would have been under alternative randomizations of treatment assignment. In *randomization inference*, introduced by Fisher (1935), the units and their potential outcomes are treated as fixed, while the assignment is random. The reference distribution of a statistic is that induced by the randomization of the treatment allocation. One takes *"a design-based perspective where the properties of the estimators arises from the stochastic nature of the treatment assignment, rather than a sampling-based or model-based perspective where these properties arise from the random*[53] *sampling of units from a large population in combination with assumptions on this population distribution"* (Athey and Imbens, 2022). One asks: *What value would the statistic have taken under another treatment assignment generated by the same design?*

Application to hypothesis testing Both sampling-based and design-based inference follow the same broad logic. We formulate a null hypothesis H_0, choose a test statistic T, and derive its exact or approximate distribution under H_0. The p-value is the probability, under H_0, of obtaining a statistic at least as extreme as T_{obs}. A small p-value is evidence against H_0.

Fisher randomization inference generally begins with the *sharp* null hypothesis of no effect for any unit:[54]

$$H_0: \quad Y_i^1 = Y_i^0, \qquad \forall i.$$

Under the sharp null:

- If there is no effect for any unit, then each observed outcome reveals both potential outcomes: $Y_i^1 = Y_i^0 = Y_i^{\text{obs}}$. Under H_0, our data therefore represent the exact value of all the missing potential outcomes.
- We choose a statistic T (such as the regression coefficient itself), and calculate it under every possible random assignment permitted by the design. The resulting distribution of T is therefore *the* randomization distribution of the statistic under H_0.
- Under complete randomization, all assignments in the randomization set $\mathcal{D}$ are equally likely, hence the p-value for a one-sided test is
$$p = \sum_{d \in \mathcal{D}} \Pr(D = d)\, \mathbb{1}\{T(d) \geq T_{\text{obs}}\}.$$

In practice: simulation When the randomization set is sufficiently small, all possible random assignments can be enumerated, producing an exact finite-sample reference distribution and p-value. More commonly, the set is too large, so it is approximated by drawing many assignments from the original assignment mechanism, which generates approximate p-values. We repeat a large number of times, e.g. $B = 10{,}000$, the following procedure, to construct the randomization distribution of the statistic under H_0:

1. Draw a treatment assignment from the randomization set[55], respecting the structure of the original assignment mechanism (e.g., within strata), thereby generating fake treatment statuses.
2. Recalculate the test statistic using this assignment. For instance, estimate the regression model if the test statistic is the slope coefficient β itself.

[53] Note that both perspectives use hypothetical repetitions, but the source of randomness differs: in sampling-based inference, the sample is random; in randomization inference, the treatment assignment is randomized.

[54] Fisher's *sharp* null differs from Neyman's *weak* null of no average treatment effect. Note that this is a distinction between hypotheses, not inherently between randomization- and sampling-based inference.

[55] (Rubin, 1974) defines the *randomization set* as *"the set of allocations that were equally likely to be observed given the randomization plan."* For example, in a completely randomized experiment with $2N$ units and exactly N assigned to each arm, there are $\binom{2N}{N}$ possible allocations.

Conventional sampling-based inference	Fisher randomization inference
Source of repeated variation	
Random samples from a population.	Treatment assignments generated by the known assignment mechanism.
Typical H_0	
Weak null of no average effect: $H_0 : \tau = \mathbb{E}[Y_i^1 - Y_i^0] = 0$	Sharp null of no effect for any unit: $H_0 : Y_i^1 - Y_i^0 = 0, \forall i$
Distribution of T under H_0	
$T := \frac{\hat{\beta} - \beta}{\mathrm{SE}[\hat{\beta}]} = \frac{\hat{\beta} - 0}{\mathrm{SE}[\hat{\beta}]} \underset{H_0}{\overset{a}{\sim}} \mathcal{N}(0,1)$ Typically, the asymptotic sampling distribution of T (across all random samples) under H_0 is our reference distribution.	$T := \hat{\beta}$ The exact or approximate distribution of T (across all or 10,000 random assignments) is our reference distribution.
Two-sided p-value $=$ Pr $\left[\text{observing a } \lvert T \rvert > \lvert T_{\text{obs}} \rvert \mid H_0\right]$ $=$ share of the distribution that is $> \lvert T_{\text{obs}} \rvert$	
$:= \Pr\big($*obtaining a difference between groups as extreme as observed*$\big)$ if they had been drawn from underlying sampling frames with no mean difference.	$:= \Pr\big($*obtaining a difference between groups as extreme as observed*$\big)$ if the treatment effect were in fact zero for every subject.
$\Longrightarrow$ Given e.g. a rejection threshold $\alpha = 0.05$, the α-level test will erroneously reject H_0 less than 5% of the time	

Why choose randomization-based inference instead of sampling-based inference?

- *Finite-population inference:* If all units in the target population are observed, there may be no relevant sampling uncertainty, but causal uncertainty remains because only one treatment assignment was realized.[56]
- *Finite-sample validity and weak distributional assumptions:* Under a sharp null and a known assignment mechanism, RI does not rely on large-sample approximations. It can therefore be useful with small samples. It does not rely on assumptions such as normally distributed outcomes either, and so easily accommodates outcomes such as counts, durations or ranks (Gerber and Green, 2012, p.63). Note that exactness does not imply high power: a small randomization set will produce coarse p-values...
- *Particular clustered designs:* The randomization distribution can directly reproduce cluster-level assignment.
  - *Small number of assignment clusters:* When the number of clusters is small, conventional cluster-robust asymptotic inference may be unreliable. RI can instead use the exact cluster-level assignment mechanism, and hence remains valid.
  - *Complex spatially correlated assignments:* If the assignment clustering isn't within well-defined boundaries, one can't rely on common methods to estimate correct standard errors (clusters can't be defined; other sandwich-type covariance matrix estimators require additional modeling assumptions...). Ex: weather variables such as rainfall are often used as a strategy for causal inference, as rainfall shocks are as-if randomly assigned. However, the assignment of rainfall is highly correlated across space in an unformalizable structure. Cooperman (2017) uses national draws of historical rainfall patterns as potential assignments, thereby preserving their spatial dependence without specifying a parametric spatial covariance model. Validity requires the historical fields to represent the relevant rainfall assignment process.[57]
- *High-leverage observations:* Young (2019) shows that conventional robust and clustered inference can suffer substantial size distortions in the presence of leverage concentrated in a few observations or clusters. In his reanalysis of 53 experimental papers, randomization tests produced 13–22% fewer

[56] If all units in the finite target population are observed, there is no sampling uncertainty for descriptive parameters of that population. Conceptually, there is no true sampling variation to speak of, making sampling-based p-values meaningless. One may instead define a larger superpopulation, and consider the universe of values that units could have taken, but this changes the target of inference.

[57] Using historical rainfall fields as possible assignments requires them to be exchangeable with the rainfall process generating the study-year assignment. Climate change or other trends may violate this stationarity assumption.

individually significant treatment effects than the authors' methods.[58] This does not mean that RI is generally insensitive to influential observations: the chosen statistic itself may remain highly sensitive to them.

Limitations

- RI requires a known or credibly specified assignment mechanism. It cannot generally be implemented merely by arbitrarily permuting treatment labels.
- Classical Fisher RI produces finite-sample exact tests when the null hypothesis is sufficiently *sharp* to impute all missing potential outcomes. The weak null $H_0 : \frac{1}{N}\sum_i(Y_i^1 - Y_i^0) = 0$ does not do so when effects are heterogeneous. It is therefore not adequate to test hypotheses such as a null average treatment effect.
- Because RI generally conditions on the units in the experiment, it does not by itself quantify sampling uncertainty or establish external validity beyond those units.
- A single reference distribution under $H_0(0)$ does not by itself provide a confidence interval for $\hat{\beta}$. It describes how the statistic would vary across assignments if the specified null were true, not our statistical uncertainty over every possible value of the causal effect. RI may also be used for construction of confidence intervals, but this requires additional assumptions.
  - ↪ *It is legitimate to ask whether RI is even worth it then, given the heavy criticism of the two-way binary approach to statistical hypothesis testing, based on the NHST falsificationist paradigm and the formulation of binary statements of 'statistical significance' from a p-value threshold...*
  - – Rosenbaum (2002, p.45) constructs confidence intervals by inverting a family of randomization tests. For example, testing each sharp constant-effect null: $H_0(c) : Y_i^1 - Y_i^0 = c$, $\forall i$, and retaining the values of c that are not rejected produces an exact confidence set for the common additive effect. Its interpretation, however, relies on the homogeneous-effect assumption.
  - – Gerber and Green (2012, p.67) proposes a simpler—but less accurate—method, and argues that the two methods tend to produce similar results, especially in large samples.
  - – Barrios et al. (2012, eq. 4.2) gives the exact conditional (randomization-based) variance of $\hat{\beta}$ (in the univariate linear regression model of Y on D) under the assumption of a homogeneous treatment effect, based on Neyman (1923):

$$\mathbb{V}_R[\hat{\beta}_{\text{OLS}} \mid Y^1, Y^0] = \frac{N}{N_1(N - N_1)(N - 1)} \sum_i (Y_i^0 - \bar{Y^0})^2, \quad \text{where } N_1 := \sum_i D_i.$$

[58] He also found that with the removal of just one observation, 35% of 0.01-significant reported results in the average paper can be rendered insignificant at that level, 16% of 0.01-insignificant reported results can be found to be significant at that level.

6.2 Structural Equation Models (SEMs)

Structural equation models are systems of equations specifying relationships among multiple observed and possibly latent variables. Under additional causal assumptions, the relationships in the network may be interpreted as direct and indirect causal effects.

SEMs are increasingly popular in ecological research. They are often represented using path diagrams. Recursive SEMs can be represented as directed acyclic graphs (DAGs), where arrows indicate directional relationships between observed variables.

Implicit assumptions – what separate SEMs from traditional modeling approaches:

1. SEMs often implicitly assume that the relationships among variables (paths) are causal. This is a big leap from the traditional statistics' "correlation does not imply causation." By using pre-existing knowledge of the system, one makes an informed hypothesis about the causal structure of the variables, and the SEM explicitly tests this supposed causal structure.
2. Variables can be both predictors and responses. A SEM is a convenient way to represent them jointly, and is thereby useful for testing and quantifying indirect (cascading) effects.

Covariance-based SEM

Estimation: Parameters are estimated jointly, often by maximum likelihood, by minimizing the discrepancy between covariance matrix of the observed sample and that implied by the model.

Goodness-of-fit: The χ^2 test evaluates the null hypothesis that the population covariance matrix equals the covariance matrix implied by the model.

Assumptions

- correctly specified and identified model;
- independent observations;
- distributional assumptions required by the estimator (e.g., multivariate normality for ML);
- residual covariances are zero unless modeled.

Limitations

- Standard SEMs are less flexible for non-normal outcomes and hierarchical, spatial, or temporal dependence, although extensions exist;
- computationally demanding (depending on the sizes of the variance-covariance matrix);
- good model fit does not validate the assumed causal structure

Piecewise SEM

Estimation: Decompose the network into its component equations and estimate each relationship separately (i.e., separate covariance matrices). The estimated paths are then combined to evaluate the entire system.

⇒ Much easier to estimate than one huge matrix → accomodate large networks more easily;
⇒ Flexible: each equation may use a different suitable model class: generalized linear models, models with spatial correlation...

Goodness-of-fit: Model fit is assessed using tests of directed separation: *are any paths missing from the model?*
Each of the q conditional-independence claims implied by the graph (i.e., the pairs without arrows in the DAG) is tested, controlling for the variables on which the given path is conditional. Fisher's statistic then aggregates their p-values $p_1, \dots, p_q$:

$$C = -2 \sum_{j=1}^{q} \ln(p_j) \overset{a}{\sim} \chi^2_{2q}.$$

A small p-value for C means that the model's conditional-independence assumptions are rejected...

Model comparison: We can compute an AIC score for the SEM.

6.3 Structural Vector Autoregression (SVAR)

See Ghanem and Smith (2021), and Greene 2002, Section 19.6.7.

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A Maths of potential outcomes

The steps that were overlooked in the main document are provided here in blue.

2.1.1 The original selection bias problem

$$\mathbb{E}[Y_i \mid D_i{=}1] - \mathbb{E}[Y_i \mid D_i{=}0] = \mathbb{E}[Y_i^1 \mid D_i{=}1] - \mathbb{E}[Y_i^0 \mid D_i{=}0] \qquad \text{(definition of potential outcomes)}$$

$$= \mathbb{E}[Y_i^1 \mid D_i{=}1] - \mathbb{E}[Y_i^0 \mid D_i{=}1] + \mathbb{E}[Y_i^0 \mid D_i{=}1] - \mathbb{E}[Y_i^0 \mid D_i{=}0]$$

$$= \underbrace{\mathbb{E}[Y_i^1 - Y_i^0 \mid D_i{=}1]}_{\text{ATET}} + \underbrace{\mathbb{E}[Y_i^0 \mid D_i{=}1] - \mathbb{E}[Y_i^0 \mid D_i{=}0]}_{\text{selection bias}}$$

The same demonstration holds conditional on X_i, i.e., within each stratum of X_i:

$$\mathbb{E}[Y_i \mid D_i{=}1, \mathrm{X}_i{=}\mathrm{x}] - \mathbb{E}[Y_i \mid D_i{=}0, \mathrm{X}_i{=}\mathrm{x}]$$

$$= \mathbb{E}[Y_i^1 \mid D_i{=}1, \mathrm{X}_i{=}\mathrm{x}] - \mathbb{E}[Y_i^0 \mid D_i{=}0, \mathrm{X}_i{=}\mathrm{x}]$$

$$= \mathbb{E}[Y_i^1 \mid D_i{=}1, \mathrm{X}_i{=}\mathrm{x}] - \mathbb{E}[Y_i^0 \mid D_i{=}1, \mathrm{X}_i{=}\mathrm{x}] + \mathbb{E}[Y_i^0 \mid D_i{=}1, \mathrm{X}_i{=}\mathrm{x}] - \mathbb{E}[Y_i^0 \mid D_i{=}0, \mathrm{X}_i{=}\mathrm{x}]$$

$$= \underbrace{\mathbb{E}[Y_i^1 - Y_i^0 \mid D_i{=}1, \mathrm{X}_i{=}\mathrm{x}]}_{\text{ATET at x}} + \underbrace{\mathbb{E}[Y_i^0 \mid D_i{=}1, \mathrm{X}_i{=}\mathrm{x}] - \mathbb{E}[Y_i^0 \mid D_i{=}0, \mathrm{X}_i{=}\mathrm{x}]}_{\text{selection bias at x}}$$

2.1.2 Average treatment effects as regression slopes and endogeneity

Simplest setting: binary D, no X

The population OLS slope estimand simplifies to the difference in average observed outcomes, which we show itself simplifies to a condition on the *causal disturbance* u_i^c:

$$\beta_{\text{OLS}} := \frac{\operatorname{cov}[Y_i, D_i]}{\mathbb{V}[D_i]} = \frac{\mathbb{E}[Y_i D_i] - \mathbb{E}[Y_i]\,\mathbb{E}[D_i]}{\mathbb{E}[D_i^2] - \mathbb{E}[D_i]^2}$$

$$= \frac{\mathbb{E}[Y_i \mid D_i{=}1]\,\Pr(D_i{=}1) - \Big(\mathbb{E}[Y_i \mid D_i{=}0]\,\Pr(D_i{=}0) + \mathbb{E}[Y_i \mid D_i{=}1]\,\Pr(D_i{=}1)\Big)\,\Pr(D_i{=}1)}{\Pr(D_i{=}1) - \Pr(D_i{=}1)^2}$$

$$= \frac{\mathbb{E}[Y_i \mid D_i{=}1]\,\Pr(D_i{=}1)\,\big(1 - \Pr(D_i{=}1)\big) - \mathbb{E}[Y_i \mid D_i{=}0]\,\Pr(D_i{=}0)\,\Pr(D_i{=}1)}{\Pr(D_i{=}1)\big(1 - \Pr(D_i{=}1)\big)}$$

$$= \frac{\mathbb{E}[Y_i \mid D_i{=}1]\,\Pr(D_i{=}1)\,\Pr(D_i{=}0) - \mathbb{E}[Y_i \mid D_i{=}0]\,\Pr(D_i{=}0)\,\Pr(D_i{=}1)}{\Pr(D_i{=}1)\,\Pr(D_i{=}0)}$$

$$= \mathbb{E}[Y_i \mid D_i{=}1] - \mathbb{E}[Y_i \mid D_i{=}0]$$

From the causal regression-form decomposition, $Y_i = \alpha^c + \beta^c D_i + u_i^c$ (where β^c is the ATET), hence:

$$\begin{cases} \mathbb{E}[Y_i | D_i{=}1] = \alpha^c + \beta^c + \mathbb{E}[u_i^c | D_i{=}1] \\ \mathbb{E}[Y_i | D_i{=}0] = \alpha^c + \mathbb{E}[u_i^c | D_i{=}0] \end{cases} \Rightarrow \quad \beta_{\text{OLS}} = \beta^c + \mathbb{E}[u_i^c | D_i{=}1] - \mathbb{E}[u_i^c | D_i{=}0]$$

We can also recover again the expression with selection bias, from the expression of the causal disturbance $u_i^c := Y_i^0 - \mathbb{E}[Y_i^0 \mid D_i{=}0] + (\beta_i - \mathbb{E}[\beta_i \mid D_i{=}1])\,D_i$:

$$\begin{cases} \mathbb{E}[u_i^c | D_i{=}1] = \mathbb{E}[Y_i^0 | D_i{=}1] - \mathbb{E}[Y_i^0 | D_i{=}0] \\ \mathbb{E}[u_i^c | D_i{=}0] = 0 \end{cases} \Rightarrow \quad \mathbb{E}[u_i^c | D_i{=}1] - \mathbb{E}[u_i^c | D_i{=}0] = \mathbb{E}[Y_i^0 | D_i{=}1] - \mathbb{E}[Y_i^0 | D_i{=}0]$$

Therefore $\beta_{\text{OLS}} = \ldots = \mathbb{E}[Y_i \mid D_i{=}1] - \mathbb{E}[Y_i \mid D_i{=}0] = \ldots = \text{ATET} + \text{selection bias}.$

With covariates X

For simplicity, consider a single discrete X_i.

- Matching estimand

$$\beta_{\text{M}} = \sum_x \delta_x \ \Pr(X_i{=}x \mid D_i{=}1) = \sum_x \delta_x \frac{\Pr(X_i{=}x, D_i{=}1)}{\Pr(D_i{=}1)} = \sum_x \delta_x \frac{\Pr(D_i{=}1|X_i{=}x)\ \Pr(X_i{=}x)}{\Pr(D_i{=}1)}$$

$$= \frac{1}{\Pr(D_i{=}1)} \sum_x \delta_x \ \Pr(D_i{=}1|X_i{=}x)\ \Pr(X_i{=}x)$$

$$= \frac{\sum_x \delta_x \ \Pr(D_i{=}1|X_i{=}x)\ \Pr(X_i{=}x)}{\sum_x \Pr(D_i{=}1|X_i{=}x)\ \Pr(X_i{=}x)}$$

- OLS estimand
  The demonstration uses the Frisch-Waugh-Lovell theorem; See Angrist and Pischke (2008, p.55).

3.1 IV

IV estimand

$$\beta_{\text{IV}} := \frac{\text{cov}[Y_i, Z_i]}{\text{cov}[D_i, Z_i]} = \frac{\mathbb{E}[Y_i Z_i] - \mathbb{E}[Y_i]\,\mathbb{E}[Z_i]}{\mathbb{E}[D_i Z_i] - \mathbb{E}[D_i]\,\mathbb{E}[Z_i]}$$

$$= \frac{\mathbb{E}[Y_i \mid Z_i{=}1]\Pr(Z_i{=}1) - \Big(\mathbb{E}[Y_i \mid Z_i{=}1]\Pr(Z_i{=}1) + \mathbb{E}[Y_i \mid Z_i{=}0]\Pr(Z_i{=}0)\Big)\Pr(Z_i{=}1)}{\mathbb{E}[D_i \mid Z_i{=}1]\Pr(Z_i{=}1) - \Big(\mathbb{E}[D_i \mid Z_i{=}1]\Pr(Z_i{=}1) + \mathbb{E}[D_i \mid Z_i{=}0]\Pr(Z_i{=}0)\Big)\Pr(Z_i{=}1)}$$

$$= \frac{\mathbb{E}[Y_i \mid Z_i{=}1]\big(1 - \Pr(Z_i{=}1)\big) - \mathbb{E}[Y_i \mid Z_i{=}0]\Pr(Z_i{=}0)}{\mathbb{E}[D_i \mid Z_i{=}1]\big(1 - \Pr(Z_i{=}1)\big) - \mathbb{E}[D_i \mid Z_i{=}0]\Pr(Z_i{=}0)}$$

$$= \frac{\mathbb{E}[Y_i \mid Z_i{=}1] - \mathbb{E}[Y_i \mid Z_i{=}0]}{\mathbb{E}[D_i \mid Z_i{=}1] - \mathbb{E}[D_i \mid Z_i{=}0]}$$

- Numerator: $\mathbb{E}\big[Y_i \mid Z_i{=}1\big] - \mathbb{E}\big[Y_i \mid Z_i{=}0\big] =$

$$= \sum_{\substack{(D_i^0,D_i^1)\\ \in\{0,1\}^2}} \mathbb{E}\left[Y_i|Z_i{=}1,D_i^0,D_i^1\right]\ \Pr\left(D_i^0,D_i^1|Z_i{=}1\right) - \sum_{\substack{(D_i^0,D_i^1)\\ \in\{0,1\}^2}} \mathbb{E}\left[Y_i|Z_i{=}0,D_i^0,D_i^1\right]\ \Pr\left(D_i^0,D_i^1|Z_i{=}0\right)$$

$$= \sum_{\substack{(D_i^0,D_i^1)\\ \in\{0,1\}^2}} \mathbb{E}\big[Y_i^{D_i^1} \mid D_i^0,D_i^1\big]\Pr\left(D_i^0,D_i^1\right) - \sum_{\substack{(D_i^0,D_i^1)\\ \in\{0,1\}^2}} \mathbb{E}\big[Y_i^{D_i^0} \mid D_i^0,D_i^1\big]\Pr\left(D_i^0,D_i^1\right)$$

$$= \sum_{(D_i^0,D_i^1)\in\{0,1\}^2} \Big(\mathbb{E}\big[Y_i^{D_i^1} \mid D_i^0,D_i^1\big] - \mathbb{E}\big[Y_i^{D_i^0} \mid D_i^0,D_i^1\big]\Big)\Pr\left(D_i^0,D_i^1\right)$$

$$= \Big(\mathbb{E}\big[Y_i^0 \mid D_i^0{=}0, D_i^1{=}0\big] - \mathbb{E}\big[Y_i^0 \mid D_i^0{=}0, D_i^1{=}0\big]\Big)\ \Pr\left(D_i^0{=}0, D_i^1{=}0\right)$$
$$+ \Big(\mathbb{E}\big[Y_i^1 \mid \text{——}\,0, \text{——}\,1\big] - \mathbb{E}\big[Y_i^0 \mid \text{——}\,0, \text{——}\,1\big]\Big)\ \Pr\big(\text{——}\,0, \text{——}\,1\big)$$
$$+ \Big(\mathbb{E}\big[Y_i^0 \mid \text{——}\,1, \text{——}\,0\big] - \mathbb{E}\big[Y_i^1 \mid \text{——}\,1, \text{——}\,0\big]\Big)\ \Pr\big(\text{——}\,1, \text{——}\,0\big)$$
$$+ \Big(\mathbb{E}\big[Y_i^1 \mid \text{——}\,1, \text{——}\,1\big] - \mathbb{E}\big[Y_i^1 \mid \text{——}\,1, \text{——}\,1\big]\Big)\ \Pr\big(\text{——}\,1, \text{——}\,1\big)$$
$$= \mathbb{E}\big[Y_i^1 - Y_i^0 \mid D_i^0{=}0, D_i^1{=}1\big]\Pr\big(D_i^0{=}0, D_i^1{=}1\big) - \mathbb{E}\big[Y_i^1 - Y_i^0 \mid D_i^0{=}1, D_i^1{=}0\big]\Pr\big(D_i^0{=}1, D_i^1{=}0\big)$$
$$= \mathbb{E}\big[Y_i^1 - Y_i^0 \mid D_i^0{=}0, D_i^1{=}1\big]\Pr\big(D_i^0{=}0, D_i^1{=}1\big), \quad \text{as the probability of defiers is 0}$$

- Denominator:
  $\mathbb{E}\big[D_i \mid Z_i{=}1\big] - \mathbb{E}\big[D_i \mid Z_i{=}0\big] = \mathbb{E}\big[D_i^1 - D_i^0\big]$

$$= 1 \times \Pr\big(D_i^1 - D_i^0 = 1\big) \ + \ 0 \times \Pr\big(D_i^1 - D_i^0 = 0\big) \ - 1 \times \Pr\big(D_i^1 - D_i^0 = -1\big)$$
$$= \Pr\big(D_i^0{=}0, D_i^1{=}1\big), \quad \text{as the probability of defiers is 0}$$

$$\Longrightarrow \frac{\mathbb{E}[Y_i \mid Z_i{=}1] - \mathbb{E}[Y_i \mid Z_i{=}0]}{\mathbb{E}[D_i \mid Z_i{=}1] - \mathbb{E}[D_i \mid Z_i{=}0]} = \underbrace{\mathbb{E}\big[Y_i^1 - Y_i^0 \mid D_i^0{=}0, D_i^1{=}1\big]}_{\textbf{LATE for the compliers}}$$

Counting and characterizing compliers to get more out of the LATE

- Size of the complier group: It is the Wald first-stage, as, given monotonicity:

$$\Pr(D_i^1 > D_i^0) = \Pr(D_i^1 - D_i^0 = 1) = \mathbb{E}[D_i^1 - D_i^0] = \mathbb{E}[D_i^1] - \mathbb{E}[D_i^0] = \mathbb{E}[D_i|Z_i{=}1] - \mathbb{E}[D_i|Z_i{=}0]$$

- Share of treated that are compliers:

$$\begin{aligned}
\Pr(D_i^1 > D_i^0 \mid D_i{=}1) &= \frac{\Pr(D_i^1 > D_i^0, D_i{=}1)}{\Pr(D_i{=}1)} = \frac{\Pr(D_i{=}1 \mid D_i^1 > D_i^0)\Pr(D_i^1 > D_i^0)}{\Pr(D_i{=}1)} \\
&= \frac{\Pr(Z_i{=}1 \mid D_i^1 > D_i^0)\Pr(D_i^1 > D_i^0)}{\Pr(D_i{=}1)} \\
&= \frac{\Pr(Z_i{=}1)\Pr(D_i^1 > D_i^0)}{\Pr(D_i{=}1)} \\
&= \frac{\Pr(Z_i{=}1) \times (\mathbb{E}[D_i|Z_i{=}1] - \mathbb{E}[D_i|Z_i{=}0])}{\Pr(D_i{=}1)} \\
&= \frac{\Pr(\text{instrument is switched on}) \times \text{1st stage}}{\text{share}_{\text{treated}}}
\end{aligned}$$

- Distribution of covariates X for compliers:
  - For binary characteristics X, we can compute relative likelihoods. First we assume also conditional independence $Z_i \perp\!\!\!\perp (D_i^0, D_i^1) \mid X_i$, which implies

$$\begin{aligned}
\mathbb{E}[D_i \mid Z_i{=}1, X_i{=}x] - \mathbb{E}[D_i \mid Z_i{=}0, X_i{=}x] &= \mathbb{E}[D_i^1 \mid Z_i{=}1, X_i{=}x] - \mathbb{E}[D_i^0 \mid Z_i{=}0, X_i{=}x] \\
&= \mathbb{E}[D_i^1 - D_i^0 \mid X_i{=}x] \quad \text{(cond. independence)} \\
&= \mathbb{E}\left[\mathbb{1}\{D_i^1 > D_i^0\} \mid X_i{=}x\right] \quad \text{(binary } D_i\text{)} \\
&= \Pr(D_i^1 > D_i^0 \mid X_i{=}x).
\end{aligned}$$

Hence:

$$\begin{aligned}
\frac{\Pr(X_i{=}1 \mid D_i^1 > D_i^0)}{\Pr(X_i{=}1)} &= \frac{\Pr(X_i{=}1, D_i^1 > D_i^0)}{\Pr(X_i{=}1)\Pr(D_i^1 > D_i^0)} = \frac{\Pr(D_i^1 > D_i^0 \mid X_i{=}1)}{\Pr(D_i^1 > D_i^0)} \\
&= \frac{\mathbb{E}[D_i \mid Z_i{=}1, X_i{=}1] - \mathbb{E}[D_i \mid Z_i{=}0, X_i{=}1]}{\mathbb{E}[D_i \mid Z_i{=}1] - \mathbb{E}[D_i \mid Z_i{=}0]} \\
&= \frac{\text{1st stage} \mid X_i{=}1}{\text{1st stage}}
\end{aligned}$$

3.2 RD

Sharp RD estimand

$$\begin{aligned}
\beta_{\text{RD}} &:= \lim_{x \to c^+} \mathbb{E}[Y_i \mid X_i{=}x] - \lim_{x \to c^-} \mathbb{E}[Y_i \mid X_i{=}x] \\
&= \lim_{x \to c^+} \mathbb{E}[Y_i^1 \mid X_i{=}x] - \lim_{x \to c^-} \mathbb{E}[Y_i^0 \mid X_i{=}x] \\
&= \mathbb{E}[Y_i^1 \mid X_i{=}c] - \mathbb{E}[Y_i^0 \mid X_i{=}c] \\
&= \underbrace{\mathbb{E}[Y_i^1 - Y_i^0 \mid X_i{=}c]}_{\textbf{LATE at the cutoff}}
\end{aligned}$$

Fuzzy RD estimand

$$\beta_{\text{RD}} := \frac{\lim_{x\to c^+}\mathbb{E}[Y_i \mid X_i{=}x] - \lim_{x\to c^-}\mathbb{E}[Y_i \mid X_i{=}x]}{\lim_{x\to c^+}\mathbb{E}[D_i \mid X_i{=}x] - \lim_{x\to c^-}\mathbb{E}[D_i \mid X_i{=}x]} = \lim_{\delta\to 0}\frac{\mathbb{E}[Y_i \mid c < X_i < c+\delta] - \mathbb{E}[Y_i \mid c-\delta < X_i < c]}{\mathbb{E}[D_i \mid c < X_i < c+\delta] - \mathbb{E}[D_i \mid c-\delta < X_i < c]}$$

$$= \frac{\mathbb{E}\left[Y_i^{D_i^1} - Y_i^{D_i^0} \mid X_i{=}c\right]}{\mathbb{E}\left[D_i^1 - D_i^0 \mid X_i{=}c\right]} = \frac{\mathbb{E}\left[(Y_i^1 - Y_i^0)(D_i^1 - D_i^0) \mid X_i{=}c\right]}{\mathbb{E}\left[D_i^1 - D_i^0 \mid X_i{=}c\right]} \qquad \text{(local continuity)}$$

$$= \frac{\mathbb{E}\left[Y_i^1 - Y_i^0 \mid D_i^1 > D_i^0, X_i{=}c\right]\Pr(D_i^1 > D_i^0 \mid X_i{=}c)}{\Pr(D_i^1 > D_i^0 \mid X_i{=}c)} \qquad \text{(local monotonicity)}$$

$$= \underbrace{\mathbb{E}\left[Y_i^1 - Y_i^0 \mid D_i^1 > D_i^0,\ X_i{=}c\right]}_{\textbf{LATE for compliers at the cutoff}}$$

3.3 DiD, DiDiD, event-studies

DiD estimand

- In a before/after comparison within the treatment group, the observed change in mean outcomes could reflect other changes occurring over the same period (time effects), rather than treatment alone:

$$\begin{aligned}\mathbb{E}[Y_{i1} \mid G_i{=}1] - \mathbb{E}[Y_{i0} \mid G_i{=}1] &= \mathbb{E}[Y_{i1}^1 - Y_{i0} - Y_{i1}^0 + Y_{i1}^0 \mid G_i{=}1]\\ &= \underbrace{\mathbb{E}[Y_{i1}^1 - Y_{i1}^0 \mid G_i{=}1]}_{\text{ATET}} + \underbrace{\mathbb{E}[Y_{i1}^0 - Y_{i0} \mid G_i{=}1]}_{\substack{\text{untreated time trend}\\ = \text{ before/after bias}}}\end{aligned}$$

- In a treatment/control (i.e., cross-sectional) comparison within the post period, the observed difference in mean outcomes could reflect permanent differences between the groups, rather than treatment alone:

$$\begin{aligned}\mathbb{E}[Y_{i1} \mid G_i{=}1] - \mathbb{E}[Y_{i1} \mid G_i{=}0] &= \mathbb{E}[Y_{i1}^1 - Y_{i1}^0 + Y_{i1}^0 \mid G_i{=}1] - \mathbb{E}[Y_{i1}^0 \mid G_i{=}0]\\ &= \underbrace{\mathbb{E}[Y_{i1}^1 - Y_{i1}^0 \mid G_i{=}1]}_{\text{ATET}} + \underbrace{\mathbb{E}[Y_{i1}^0 \mid G_i{=}1] - \mathbb{E}[Y_{i1}^0 \mid G_i{=}0]}_{\substack{\text{untreated group difference}\\ = \text{ selection bias}}}\end{aligned}$$

- The DiD estimand compares the *change over time in* mean outcomes in the treatment group to the *change over time in* mean outcomes in the control group, with a remaining bias that equals zero under parallel trends:

$$\begin{aligned}\beta_{\text{DiD}} &:= \left(\bar{Y}_{G_1P_1} - \bar{Y}_{G_1P_0}\right) - \left(\bar{Y}_{G_0P_1} - \bar{Y}_{G_0P_0}\right)\\ &:= \Big(\mathbb{E}_i\big[Y_{i1} \mid G_i{=}1\big] - \mathbb{E}_i\big[Y_{i0} \mid G_i{=}1\big]\Big) - \Big(\mathbb{E}_i\big[Y_{i1} \mid G_i{=}0\big] - \mathbb{E}_i\big[Y_{i0} \mid G_i{=}0\big]\Big)\\ &= \mathbb{E}\left[Y_{i1}^1 - Y_{i0} \mid G_i{=}1\right] - \mathbb{E}\left[Y_{i1}^0 - Y_{i0} \mid G_i{=}0\right]\\ &= \mathbb{E}\left[Y_{i1}^1 - Y_{i0} \mid G_i{=}1\right] - \mathbb{E}\left[Y_{i1}^0 - Y_{i0} \mid \boldsymbol{G_i{=}1}\right] \qquad (\parallel \text{trends})\\ &= \underbrace{\mathbb{E}\left[Y_{i1}^1 - Y_{i1}^0 \mid G_i{=}1\right]}_{\textbf{ATET in the post period}}\end{aligned}$$

DiDiD estimand

Define $\mu_{gsp} := \mathbb{E}\left[Y_{it} \mid G_i{=}g, S_i{=}s, P_t{=}p\right]$.

$$
\begin{aligned}
\beta_{\text{DiDiD}} &:= \left[(\mu_{111} - \mu_{110}) - (\mu_{011} - \mu_{010})\right] - \left[(\mu_{101} - \mu_{100}) - (\mu_{001} - \mu_{000})\right] \\
&= \left(\mathbb{E}\left[Y_{i1}{-}Y_{i0} \mid G1, S1\right] - \mathbb{E}\left[Y_{i1}{-}Y_{i0} \mid G1, S0\right]\right) - \left(\mathbb{E}\left[Y_{i1}{-}Y_{i0} \mid G0, S1\right] - \mathbb{E}\left[Y_{i1}{-}Y_{i0} \mid G0, S0\right]\right) \\
&= \left(\mathbb{E}\left[Y_{i1}^1{-}Y_{i0} \mid G1, S1\right] - \mathbb{E}\left[Y_{i1}^0{-}Y_{i0} \mid G1, S0\right]\right) - \left(\mathbb{E}\left[Y_{i1}^0{-}Y_{i0} \mid G0, S1\right] - \mathbb{E}\left[Y_{i1}^0{-}Y_{i0} \mid G0, S0\right]\right) \\
&= \left(\mathbb{E}\left[Y_{i1}^1{-}Y_{i0} \mid G1, S1\right] - \mathbb{E}\left[Y_{i1}^0{-}Y_{i0} \mid G1, S0\right]\right) - \left(\mathbb{E}\left[Y_{i1}^0{-}Y_{i0} \mid \boldsymbol{G1, S1}\right] - \mathbb{E}\left[Y_{i1}^0{-}Y_{i0} \mid \boldsymbol{G1, S0}\right]\right) \\
&= \mathbb{E}\left[Y_{i1}^1{-}Y_{i0} \mid G1, S1\right] - \mathbb{E}\left[Y_{i1}^0{-}Y_{i0} \mid G1, S1\right] \\
&= \underbrace{\mathbb{E}\left[Y_{i1}^1 - Y_{i1}^0 \mid G_i{=}1, S_i{=}1\right]}_{\textbf{ATET in the post period}}
\end{aligned}
$$